Solve Lambda Vacuum Field Equations

Case 6D_1C : 4D spacetime (t,r,θ,ϕ) + two individually curled axis (ψ)
1: Curled geometries: Circles
C: Curled metric $-\mathcal{D}_4[r]\, p_4{}^2\, d\psi^2 - \mathcal{D}_5[r]\, p_5{}^2\, d\psi^2$
3-space metric: $-B(t,r)\, dr^2 - \mathbb{C}(t,r)\, r^2\, d\theta^2 - \mathbb{C}(t,r)\, r^2 \sin^2\theta\, d\phi^2$
Time metric: $A(t,r)\, dt^2$

Line Element

Out[ ]=

$$ds^2 = A\, dt^2 - B\, dr^2 - r^2\, \mathbb{C}\, d\theta^2 - r^2\, \mathbb{C}\, \mathrm{Sin}[\theta]^2\, d\phi^2 - p_4^2\, \mathcal{D}_4\, d\psi_4^2 - p_5^2\, \mathcal{D}_5\, d\psi_5^2$$

Metric Tensor Matrices

Out[ ]=

$$g:\ (g_{\mu\nu}) = \begin{pmatrix} \frac{A}{c^2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -B & 0 & 0 & 0 & 0 \\ 0 & 0 & -r^2\,\mathbb{C} & 0 & 0 & 0 \\ 0 & 0 & 0 & -r^2\,\mathbb{C}\,\mathrm{Sin}[\theta]^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & -p_4^2\,\mathcal{D}_4 & 0 \\ 0 & 0 & 0 & 0 & 0 & -p_5^2\,\mathcal{D}_5 \end{pmatrix}$$

$$G:\ (g^{\mu\nu}) = \begin{pmatrix} \frac{c^2}{A} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{B} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{r^2\,\mathbb{C}} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\mathrm{Csc}[\theta]^2}{r^2\,\mathbb{C}} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{p_4^2\,\mathcal{D}_4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{p_5^2\,\mathcal{D}_5} \end{pmatrix}$$

Case 6D_1C: 6D = 4D spacetime (t,r,θ,ϕ) + two individually curled axis (ψ_4,ψ_5)
$g_{00}=\frac{A(t,r)}{c^2}$, $g_{11}=-B(t,r)$, $g_{22}=-\mathbb{C}(t,r)r^2$, $g_{33}=-\mathbb{C}(t,r)r^2\sin^2\theta$, $g_{44}=-\mathcal{D}_4(r)p_4{}^2$, $g_{55}=-\mathcal{D}_5(r)p_5{}^2$

We wish to be explicit upfront that in Case 1C **we assume $\mathcal{D}_4 \neq \mathcal{D}_5$** because $\mathcal{D}_4=\mathcal{D}_5$ is handled in Case 1E. Thus, as we solve equations, we do not consider setting $\mathcal{D}_4=\mathcal{D}_5$ even if it might be helpful.

The Lambda field equations can be represented as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$:

$$R_{00}: \quad -\frac{c^2 \dot{A} \dot{B}}{4 A^2 B} - \frac{c^2 \dot{B}^2}{4 A B^2} + \frac{c^2 \ddot{B}}{2 A B} - \frac{c^2 \dot{A} \dot{\mathbb{C}}}{2 A^2 \mathbb{C}} - \frac{c^2 \dot{\mathbb{C}}^2}{2 A \mathbb{C}^2} + \frac{c^2 \ddot{\mathbb{C}}}{A \mathbb{C}} - \frac{A'}{A B r} + \frac{(A')^2}{4 A^2 B} + \frac{A' B'}{4 A B^2} - \frac{A' \mathbb{C}'}{2 A B \mathbb{C}} - \frac{A' \mathbb{D}_4'}{4 A B \mathbb{D}_4} - \frac{A' \mathbb{D}_5'}{4 A B \mathbb{D}_5} - \frac{A''}{2 A B} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{B}}{B r} + \frac{\dot{\mathbb{C}}}{r \mathbb{C}} - \frac{\dot{\mathbb{C}} A'}{2 A \mathbb{C}} - \frac{\dot{B} \mathbb{C}'}{2 B \mathbb{C}} - \frac{\dot{\mathbb{C}} \mathbb{C}'}{2 \mathbb{C}^2} + \frac{\dot{\mathbb{C}}'}{\mathbb{C}} - \frac{\dot{B} \mathbb{D}_4'}{4 B \mathbb{D}_4} - \frac{\dot{B} \mathbb{D}_5'}{4 B \mathbb{D}_5} = 0$$

$$R_{11}: \quad -\frac{c^2 \dot{A} \dot{B}}{4 A^2 B} - \frac{c^2 \dot{B}^2}{4 A B^2} + \frac{c^2 \ddot{B}}{2 A B} + \frac{c^2 \dot{B} \dot{\mathbb{C}}}{2 A B \mathbb{C}} + \frac{(A')^2}{4 A^2 B} + \frac{B'}{B^2 r} + \frac{A' B'}{4 A B^2} - \frac{2 \mathbb{C}'}{B r \mathbb{C}} + \frac{B' \mathbb{C}'}{2 B^2 \mathbb{C}} + \frac{(\mathbb{C}')^2}{2 B \mathbb{C}^2} + \frac{B' \mathbb{D}_4'}{4 B^2 \mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4 B \mathbb{D}_4^2} + \frac{B' \mathbb{D}_5'}{4 B^2 \mathbb{D}_5} + \frac{(\mathbb{D}_5')^2}{4 B \mathbb{D}_5^2} - \frac{A''}{2 A B} - \frac{\mathbb{C}''}{B \mathbb{C}} - \frac{\mathbb{D}_4''}{2 B \mathbb{D}_4} - \frac{\mathbb{D}_5''}{2 B \mathbb{D}_5} = \Lambda$$

$$R_{22}: \quad -\frac{1}{B r^2} + \frac{1}{r^2 \mathbb{C}} - \frac{c^2 \dot{A} \dot{\mathbb{C}}}{4 A^2 \mathbb{C}} + \frac{c^2 \dot{B} \dot{\mathbb{C}}}{4 A B \mathbb{C}} + \frac{c^2 \ddot{\mathbb{C}}}{2 A \mathbb{C}} - \frac{A'}{2 A B r} + \frac{B'}{2 B^2 r} - \frac{2 \mathbb{C}'}{B r \mathbb{C}} - \frac{A' \mathbb{C}'}{4 A B \mathbb{C}} + \frac{B' \mathbb{C}'}{4 B^2 \mathbb{C}} - \frac{\mathbb{D}_4'}{2 B r \mathbb{D}_4} - \frac{\mathbb{C}' \mathbb{D}_4'}{4 B \mathbb{C} \mathbb{D}_4} - \frac{\mathbb{D}_5'}{2 B r \mathbb{D}_5} - \frac{\mathbb{C}' \mathbb{D}_5'}{4 B \mathbb{C} \mathbb{D}_5} - \frac{\mathbb{C}''}{2 B \mathbb{C}} = \Lambda$$

$$R_{44}: \quad -\frac{\mathbb{D}_4'}{B r \mathbb{D}_4} - \frac{A' \mathbb{D}_4'}{4 A B \mathbb{D}_4} + \frac{B' \mathbb{D}_4'}{4 B^2 \mathbb{D}_4} - \frac{\mathbb{C}' \mathbb{D}_4'}{2 B \mathbb{C} \mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4 B \mathbb{D}_4^2} - \frac{\mathbb{D}_4' \mathbb{D}_5'}{4 B \mathbb{D}_4 \mathbb{D}_5} - \frac{\mathbb{D}_4''}{2 B \mathbb{D}_4} = \Lambda$$

$$R_{55}: \quad -\frac{\mathbb{D}_5'}{B r \mathbb{D}_5} - \frac{A' \mathbb{D}_5'}{4 A B \mathbb{D}_5} + \frac{B' \mathbb{D}_5'}{4 B^2 \mathbb{D}_5} - \frac{\mathbb{C}' \mathbb{D}_5'}{2 B \mathbb{C} \mathbb{D}_5} - \frac{\mathbb{D}_4' \mathbb{D}_5'}{4 B \mathbb{D}_4 \mathbb{D}_5} + \frac{(\mathbb{D}_5')^2}{4 B \mathbb{D}_5^2} - \frac{\mathbb{D}_5''}{2 B \mathbb{D}_5} = \Lambda$$

We observed in Notebook 6D_1 that R_{33} is redundant
and, so, R_{33} is omitted here and in all of the 6D notebooks.

a: $B(t,r)=a(t)^2 \mathbb{B}(r)$ and $\mathbb{C}(t,r)=a(t)^2 \mathbb{C}(r)$

The field equations become

R_{00}:

$$\frac{3 c^2 \ddot{a}}{a A} - \frac{3 c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{A'}{a^2 A r \mathbb{B}} + \frac{(A')^2}{4 a^2 A^2 \mathbb{B}} - \frac{A' \mathbb{C}'}{2 a^2 A \mathbb{C} \mathbb{B}} + \frac{A' \mathbb{B}'}{4 a^2 A \mathbb{B}^2} - \frac{A' \mathbb{D}_4'}{4 a^2 A \mathbb{B} \mathbb{D}_4} - \frac{A' \mathbb{D}_5'}{4 a^2 A \mathbb{B} \mathbb{D}_5} - \frac{A''}{2 a^2 A \mathbb{B}} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{a} A'}{a A} - \frac{\dot{a} \mathbb{D}_4'}{2 a \mathbb{D}_4} - \frac{\dot{a} \mathbb{D}_5'}{2 a \mathbb{D}_5} = 0$$

$$R_{11}: \quad \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} + \frac{(A')^2}{4 a^2 A^2 \mathbb{B}} - \frac{2 \mathbb{C}'}{a^2 r \mathbb{C} \mathbb{B}} + \frac{(\mathbb{C}')^2}{2 a^2 \mathbb{C}^2 \mathbb{B}} + \frac{\mathbb{B}'}{a^2 r \mathbb{B}^2} + \frac{A' \mathbb{B}'}{4 a^2 A \mathbb{B}^2} + \frac{\mathbb{C}' \mathbb{B}'}{2 a^2 \mathbb{C} \mathbb{B}^2} + \frac{\mathbb{B}' \mathbb{D}_4'}{4 a^2 \mathbb{B}^2 \mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4 a^2 \mathbb{B} \mathbb{D}_4^2} + \frac{\mathbb{B}' \mathbb{D}_5'}{4 a^2 \mathbb{B}^2 \mathbb{D}_5} + \frac{(\mathbb{D}_5')^2}{4 a^2 \mathbb{B} \mathbb{D}_5^2} - \frac{A''}{2 a^2 A \mathbb{B}} - \frac{\mathbb{C}''}{a^2 \mathbb{C} \mathbb{B}} - \frac{\mathbb{D}_4''}{2 a^2 \mathbb{B} \mathbb{D}_4} - \frac{\mathbb{D}_5''}{2 a^2 \mathbb{B} \mathbb{D}_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2 \mathbb{C}} - \frac{1}{a^2 r^2 \mathbb{B}} + \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{A'}{2 a^2 A r \mathbb{B}} - \frac{2 \mathbb{C}'}{a^2 r \mathbb{C} \mathbb{B}} - \frac{A' \mathbb{C}'}{4 a^2 A \mathbb{C} \mathbb{B}} + \frac{\mathbb{B}'}{2 a^2 r \mathbb{B}^2} + \frac{\mathbb{C}' \mathbb{B}'}{4 a^2 \mathbb{C} \mathbb{B}^2} - \frac{\mathbb{D}_4'}{2 a^2 r \mathbb{B} \mathbb{D}_4} - \frac{\mathbb{C}' \mathbb{D}_4'}{4 a^2 \mathbb{C} \mathbb{B} \mathbb{D}_4} - \frac{\mathbb{D}_5'}{2 a^2 r \mathbb{B} \mathbb{D}_5} - \frac{\mathbb{C}' \mathbb{D}_5'}{4 a^2 \mathbb{C} \mathbb{B} \mathbb{D}_5} - \frac{\mathbb{C}''}{2 a^2 \mathbb{C} \mathbb{B}} = \Lambda$$

$$R_{44}: \quad -\frac{\mathbb{D}_4'}{a^2 r \mathbb{B} \mathbb{D}_4} - \frac{A' \mathbb{D}_4'}{4 a^2 A \mathbb{B} \mathbb{D}_4} - \frac{\mathbb{C}' \mathbb{D}_4'}{2 a^2 \mathbb{C} \mathbb{B} \mathbb{D}_4} + \frac{\mathbb{B}' \mathbb{D}_4'}{4 a^2 \mathbb{B}^2 \mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4 a^2 \mathbb{B} \mathbb{D}_4^2} - \frac{\mathbb{D}_4' \mathbb{D}_5'}{4 a^2 \mathbb{B} \mathbb{D}_4 \mathbb{D}_5} - \frac{\mathbb{D}_4''}{2 a^2 \mathbb{B} \mathbb{D}_4} = \Lambda$$

$$R_{55}: \quad -\frac{\mathcal{D}_5'}{a^2\, r\, B\, \mathcal{D}_5} - \frac{A'\, \mathcal{D}_5'}{4\, a^2\, A\, B\, \mathcal{D}_5} - \frac{C'\, \mathcal{D}_5'}{2\, a^2\, C\, B\, \mathcal{D}_5} + \frac{B'\, \mathcal{D}_5'}{4\, a^2\, B^2\, \mathcal{D}_5} - \frac{\mathcal{D}_4'\, \mathcal{D}_5'}{4\, a^2\, B\, \mathcal{D}_4\, \mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4\, a^2\, B\, \mathcal{D}_5^2} - \frac{\mathcal{D}_5''}{2\, a^2\, B\, \mathcal{D}_5} = \Lambda$$

Because $a(t) \neq$constant, $\dot{a}(t) \neq 0$, and so R_{01} can be simplied and solved by factoring out $\frac{\dot{a}}{a}$:

$$R_{01}: \quad -\frac{A'}{A} - \frac{\mathcal{D}_4'}{2\, \mathcal{D}_4} - \frac{\mathcal{D}_5'}{2\, \mathcal{D}_5} = 0$$

$$A(t,r) = \frac{\mathcal{A}[t]}{\sqrt{\mathcal{D}_4[r]}\, \sqrt{\mathcal{D}_5[r]}}$$ where $\mathcal{A}(t)$ is a new unknown.

R_{01} is eliminated, and the remaining equations become

$$R_{00}: \quad \frac{3\, c^2\, \ddot{a}\, \sqrt{\mathcal{D}_4}\, \sqrt{\mathcal{D}_5}}{a\, \mathcal{A}} - \frac{3\, c^2\, \dot{a}\, \dot{\mathcal{A}}\, \sqrt{\mathcal{D}_4}\, \sqrt{\mathcal{D}_5}}{2\, a\, \mathcal{A}^2} + \frac{\mathcal{D}_4'}{2\, a^2\, r\, B\, \mathcal{D}_4} + \frac{C'\, \mathcal{D}_4'}{4\, a^2\, C\, B\, \mathcal{D}_4} - \frac{B'\, \mathcal{D}_4'}{8\, a^2\, B^2\, \mathcal{D}_4} - \frac{3\, (\mathcal{D}_4')^2}{16\, a^2\, B\, \mathcal{D}_4^2} + \frac{\mathcal{D}_5'}{2\, a^2\, r\, B\, \mathcal{D}_5} + \frac{C'\, \mathcal{D}_5'}{4\, a^2\, C\, B\, \mathcal{D}_5} - \frac{B'\, \mathcal{D}_5'}{8\, a^2\, B^2\, \mathcal{D}_5} + \frac{\mathcal{D}_4'\, \mathcal{D}_5'}{8\, a^2\, B\, \mathcal{D}_4\, \mathcal{D}_5} - \frac{3\, (\mathcal{D}_5')^2}{16\, a^2\, B\, \mathcal{D}_5^2} + \frac{\mathcal{D}_4''}{4\, a^2\, B\, \mathcal{D}_4} + \frac{\mathcal{D}_5''}{4\, a^2\, B\, \mathcal{D}_5} = \Lambda$$

$$R_{11}: \quad \frac{2\, c^2\, \dot{a}^2\, \sqrt{\mathcal{D}_4}\, \sqrt{\mathcal{D}_5}}{a^2\, \mathcal{A}} + \frac{c^2\, \ddot{a}\, \sqrt{\mathcal{D}_4}\, \sqrt{\mathcal{D}_5}}{a\, \mathcal{A}} - \frac{c^2\, \dot{a}\, \dot{\mathcal{A}}\, \sqrt{\mathcal{D}_4}\, \sqrt{\mathcal{D}_5}}{2\, a\, \mathcal{A}^2} - \frac{2\, C'}{a^2\, r\, C\, B} + \frac{(C')^2}{2\, a^2\, C^2\, B} + \frac{B'}{a^2\, r\, B^2} + \frac{C'\, B'}{2\, a^2\, C\, B^2} + \frac{B'\, \mathcal{D}_4'}{8\, a^2\, B^2\, \mathcal{D}_4} - \frac{(\mathcal{D}_4')^2}{16\, a^2\, B\, \mathcal{D}_4^2} + \frac{B'\, \mathcal{D}_5'}{8\, a^2\, B^2\, \mathcal{D}_5} - \frac{\mathcal{D}_4'\, \mathcal{D}_5'}{8\, a^2\, B\, \mathcal{D}_4\, \mathcal{D}_5} - \frac{(\mathcal{D}_5')^2}{16\, a^2\, B\, \mathcal{D}_5^2} - \frac{C''}{a^2\, C\, B} - \frac{\mathcal{D}_4''}{4\, a^2\, B\, \mathcal{D}_4} - \frac{\mathcal{D}_5''}{4\, a^2\, B\, \mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\, r^2\, C} - \frac{1}{a^2\, r^2\, B} + \frac{2\, c^2\, \dot{a}^2\, \sqrt{\mathcal{D}_4}\, \sqrt{\mathcal{D}_5}}{a^2\, \mathcal{A}} + \frac{c^2\, \ddot{a}\, \sqrt{\mathcal{D}_4}\, \sqrt{\mathcal{D}_5}}{a\, \mathcal{A}} - \frac{c^2\, \dot{a}\, \dot{\mathcal{A}}\, \sqrt{\mathcal{D}_4}\, \sqrt{\mathcal{D}_5}}{2\, a\, \mathcal{A}^2} - \frac{2\, C'}{a^2\, r\, C\, B} + \frac{B'}{2\, a^2\, r\, B^2} + \frac{C'\, B'}{4\, a^2\, C\, B^2} - \frac{\mathcal{D}_4'}{4\, a^2\, r\, B\, \mathcal{D}_4} - \frac{C'\, \mathcal{D}_4'}{8\, a^2\, C\, B\, \mathcal{D}_4} - \frac{\mathcal{D}_5'}{4\, a^2\, r\, B\, \mathcal{D}_5} - \frac{C'\, \mathcal{D}_5'}{8\, a^2\, C\, B\, \mathcal{D}_5} - \frac{C''}{2\, a^2\, C\, B} = \Lambda$$

$$R_{44}: \quad -\frac{\mathcal{D}_4'}{a^2\, r\, B\, \mathcal{D}_4} - \frac{C'\, \mathcal{D}_4'}{2\, a^2\, C\, B\, \mathcal{D}_4} + \frac{B'\, \mathcal{D}_4'}{4\, a^2\, B^2\, \mathcal{D}_4} + \frac{3\, (\mathcal{D}_4')^2}{8\, a^2\, B\, \mathcal{D}_4^2} - \frac{\mathcal{D}_4'\, \mathcal{D}_5'}{8\, a^2\, B\, \mathcal{D}_4\, \mathcal{D}_5} - \frac{\mathcal{D}_4''}{2\, a^2\, B\, \mathcal{D}_4} = \Lambda$$

$$R_{55}: \quad -\frac{\mathcal{D}_5'}{a^2\, r\, B\, \mathcal{D}_5} - \frac{C'\, \mathcal{D}_5'}{2\, a^2\, C\, B\, \mathcal{D}_5} + \frac{B'\, \mathcal{D}_5'}{4\, a^2\, B^2\, \mathcal{D}_5} - \frac{\mathcal{D}_4'\, \mathcal{D}_5'}{8\, a^2\, B\, \mathcal{D}_4\, \mathcal{D}_5} + \frac{3\, (\mathcal{D}_5')^2}{8\, a^2\, B\, \mathcal{D}_5^2} - \frac{\mathcal{D}_5''}{2\, a^2\, B\, \mathcal{D}_5} = \Lambda$$

We can algebraically solve $2R_{00}+R_{44}+R_{55}$ for $\mathcal{D}_5$:

$$2R_{00}+R_{44}+R_{55}: \quad \frac{6\, c^2\, \ddot{a}\, \sqrt{\mathcal{D}_4}\, \sqrt{\mathcal{D}_5}}{a\, \mathcal{A}} - \frac{3\, c^2\, \dot{a}\, \dot{\mathcal{A}}\, \sqrt{\mathcal{D}_4}\, \sqrt{\mathcal{D}_5}}{a\, \mathcal{A}^2} = 4\, \Lambda$$

$$\mathcal{D}_5(r) = \frac{16\, \Lambda^2\, a[t]^2\, \mathcal{A}[t]^4}{9\, c^4\, \left(-2\, \mathcal{A}[t]\, \ddot{a}[t] + \dot{a}[t]\, \dot{\mathcal{A}}[t]\right)^2\, \mathcal{D}_4[r]}$$

In order for this to strictly be a function of r, the t terms would all have to cancel out, leaving

$$\mathcal{D}_5(r) = \frac{K}{\mathcal{D}_4[r]}$$ where K is a constant

We assume for now that the terms cancel out and use this fact to help us find a formula for $\mathcal{A}(t)$, which we will plug into the $\mathcal{D}_5(r)$ formula above in order to either confirm or reject the assumption.

Plugging in $\mathcal{D}_5(r) = \frac{K}{\mathcal{D}_4[r]}$, the equations become

R_{00}: $$\frac{3 c^2 \sqrt{K}\, \ddot{a}}{a\, \mathcal{A}} - \frac{3 c^2 \sqrt{K}\, \dot{a}\, \dot{\mathcal{A}}}{2 a\, \mathcal{A}^2} = \Lambda$$

R_{11}:
$$\frac{2 c^2 \sqrt{K}\, \dot{a}^2}{a^2\, \mathcal{A}} + \frac{c^2 \sqrt{K}\, \ddot{a}}{a\, \mathcal{A}} - \frac{c^2 \sqrt{K}\, \dot{a}\, \dot{\mathcal{A}}}{2 a\, \mathcal{A}^2} - \frac{2 C'}{a^2 r C B} + \frac{(C')^2}{2 a^2 C^2 B} + \frac{B'}{a^2 r B^2} + \frac{C' B'}{2 a^2 C B^2} - \frac{(\mathcal{D}_4')^2}{2 a^2 B \mathcal{D}_4^2} - \frac{C''}{a^2 C B} = \Lambda$$

R_{22}:
$$\frac{1}{a^2 r^2 C} - \frac{1}{a^2 r^2 B} + \frac{2 c^2 \sqrt{K}\, \dot{a}^2}{a^2\, \mathcal{A}} + \frac{c^2 \sqrt{K}\, \ddot{a}}{a\, \mathcal{A}} - \frac{c^2 \sqrt{K}\, \dot{a}\, \dot{\mathcal{A}}}{2 a\, \mathcal{A}^2} - \frac{2 C'}{a^2 r C B} + \frac{B'}{2 a^2 r B^2} + \frac{C' B'}{4 a^2 C B^2} - \frac{C''}{2 a^2 C B} = \Lambda$$

R_{44}: $$-\frac{\mathcal{D}_4'}{a^2 r B \mathcal{D}_4} - \frac{C' \mathcal{D}_4'}{2 a^2 C B \mathcal{D}_4} + \frac{B' \mathcal{D}_4'}{4 a^2 B^2 \mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{2 a^2 B \mathcal{D}_4^2} - \frac{\mathcal{D}_4''}{2 a^2 B \mathcal{D}_4} = \Lambda$$

R_{55}: $$\frac{\mathcal{D}_4'}{a^2 r B \mathcal{D}_4} + \frac{C' \mathcal{D}_4'}{2 a^2 C B \mathcal{D}_4} - \frac{B' \mathcal{D}_4'}{4 a^2 B^2 \mathcal{D}_4} - \frac{(\mathcal{D}_4')^2}{2 a^2 B \mathcal{D}_4^2} + \frac{\mathcal{D}_4''}{2 a^2 B \mathcal{D}_4} = \Lambda$$

R_{55} has become redundant, the same as R_{44}.

We can solve R_{00} for $\mathcal{A}$:

$$\mathcal{A}(t) = \frac{3 c^2 K\, \dot{a}[t]^2}{\alpha + \sqrt{K}\, \Lambda\, a[t]^2}$$ where α is a constant of integration.

We now plug $\mathcal{A}(t)$ into $\mathcal{D}_5(r)$, above, to determine whether or not the t-terms in the numerator have canceled with the t terms in the denominator:

$$\mathcal{D}_5(r) = \frac{144\, a^2 c^4 K^4 \Lambda^2 \dot{a}^8}{\left(\alpha + a^2 \sqrt{K}\, \Lambda\right)^4 \left(-\frac{6 c^2 K \dot{a}^2 \ddot{a}}{\alpha + a^2 \sqrt{K}\, \Lambda} + \dot{a}\left(-\frac{6 a c^2 K^{3/2} \Lambda \dot{a}^3}{\left(\alpha + a^2 \sqrt{K}\, \Lambda\right)^2} + \frac{6 c^2 K \dot{a} \ddot{a}}{\alpha + a^2 \sqrt{K}\, \Lambda}\right)\right)^2 \mathcal{D}_4}$$
$$= \frac{4 K}{\mathcal{D}_4[r]}$$

This shows that the t terms have canceled out, so our assumption above is valid. It is strange that the cancelation yields an unexpected factor of 4, but it is sufficient that the t terms have disappeared.

Since $g_{55} = \mathcal{D}_5(r) p_5{}^2$, we can roll $4K$ into the radius p_5. Since K can be either negative or positive, we get:

$$\mathcal{D}_5(r) = \frac{\pm 1}{\mathcal{D}_4[r]}.$$

R_{00} is eliminated, and we are down to 3 equations in the 3 unknowns B, C, and D_4:

$$R_{11}: \quad \frac{2\alpha}{3a^2} + \Lambda - \frac{2C'}{a^2 r C B} + \frac{(C')^2}{2a^2 C^2 B} + \frac{B'}{a^2 r B^2} + \frac{C'B'}{2a^2 C B^2} - \frac{(D_4')^2}{2a^2 B D_4^2} - \frac{C''}{a^2 C B} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2 C} + \frac{2\alpha}{3a^2} - \frac{1}{a^2 r^2 B} + \Lambda - \frac{2C'}{a^2 r C B} + \frac{B'}{2a^2 r B^2} + \frac{C'B'}{4a^2 C B^2} - \frac{C''}{2a^2 C B} = \Lambda$$

$$R_{44}: \quad -\frac{D_4'}{a^2 r B D_4} - \frac{C' D_4'}{2a^2 C B D_4} + \frac{B' D_4'}{4a^2 B^2 D_4} + \frac{(D_4')^2}{2a^2 B D_4^2} - \frac{D_4''}{2a^2 B D_4} = \Lambda$$

or

$$R_{11}: \quad \frac{2\alpha}{3} - \frac{2C'}{r C B} + \frac{(C')^2}{2C^2 B} + \frac{B'}{r B^2} + \frac{C'B'}{2C B^2} - \frac{(D_4')^2}{2B D_4^2} - \frac{C''}{C B} = 0$$

$$R_{22}: \quad \frac{1}{r^2 C} + \frac{2\alpha}{3} - \frac{1}{r^2 B} - \frac{2C'}{r C B} + \frac{B'}{2r B^2} + \frac{C'B'}{4C B^2} - \frac{C''}{2C B} = 0$$

$$R_{44}: \quad -\frac{D_4'}{a^2 r B D_4} - \frac{C' D_4'}{2a^2 C B D_4} + \frac{B' D_4'}{4a^2 B^2 D_4} + \frac{(D_4')^2}{2a^2 B D_4^2} - \frac{D_4''}{2a^2 B D_4} = \Lambda$$

We can algebraically solve $R_{11}-2R_{22}$ for B:

$$R_{11}-2R_{22}: \quad 12 C^2 D_4^2 - 12 C B D_4^2 - 4 r^2 C^2 \alpha B D_4^2 + 12 r C D_4^2 C' + 3 r^2 D_4^2 (C')^2 - 3 r^2 C^2 (D_4')^2 = 0$$

$$B(r) = \frac{3\left(4 C^2 D_4^2 + 4 r C D_4^2 C' + r^2 D_4^2 (C')^2 - r^2 C^2 (D_4')^2\right)}{4 C \left(3 + r^2 C \alpha\right) D_4^2}$$

Plugging this in, the equations are a page-long each, so we don't present them here. Nonetheless, the LHs terms all have very similar denominators, and RHS of $R_{11}-R_{22}$ equals zero, so we can put the LHS of $R_{11}-R_{22}$ over a common denominator and set the numerator to zero. This simplifies this equation enough that Mathematica can solve it for D_4:

$$D_4(r) = e^{\int_1^r \frac{\pm(2C+\rho C')}{C\rho\sqrt{1+3C\beta\rho^2+C^2\alpha\beta\rho^4}}\,d\rho}\,\zeta$$ where β and ζ are constants of integration.

We can roll up ζ into the radius p_4 since $g_{44}=-D_4 p_4{}^2$:

$$D_4(r) = \pm e^{\int_1^r \frac{\pm(2C+\rho C')}{C\rho\sqrt{1+3C\beta\rho^2+C^2\alpha\beta\rho^4}}\,d\rho}.$$

Amazingly, when we plug D_4 in, the page-long equations all become identities if $\Lambda=0$:

R_{11}: $\Lambda = \Lambda$

R_{22}: $\Lambda = \Lambda$

R_{44}: $0 = \Lambda$

All of the equations are now satisfied.

The remaining unknown, $\mathbb{C}(r)$, is a free parameter.

$\mathcal{A}(t)$ has been simplified:

$$\mathcal{A}(t) = \frac{3\,c^2\,\dot{a}[t]^2}{\alpha}$$

$\mathcal{A}$ can be further simplified by replacing α with $\frac{3\,c^2}{\gamma}$ where γ is a constant:

$$\mathcal{A}[t] = \gamma\,\dot{a}[t]^2.$$

$$\Rightarrow \quad A(t,r) = \frac{\mathcal{A}[t]}{\sqrt{\mathbb{D}_4[r]}\,\sqrt{\mathbb{D}_5[r]}} = \mathcal{A}[t] = \gamma\,\dot{a}[t]^2$$

This substitution of α also affects $\mathbb{B}$ and $\mathbb{D}_4$:

$$\mathbb{B}(r) = \frac{3\,r^2\,\beta\,\gamma\,(2\,\mathbb{C}[r] + r\,\mathbb{C}'[r])^2}{4\left(\gamma + 3\,r^2\,\beta\,\gamma\,\mathbb{C}[r] + 3\,c^2\,r^4\,\beta\,\mathbb{C}[r]^2\right)}$$

$$\mathbb{D}_4(r) = \pm e^{\int_1^r \frac{\pm(2\,\mathbb{C}+\rho\,\mathbb{C}')}{\mathbb{C}\,\rho\,\sqrt{1+3\,\mathbb{C}\,\beta\,\rho^2+\frac{3\,c^2\,\mathbb{C}^2\,\beta\,\rho^4}{\gamma}}}\,d\rho}$$

This exact solution to the Λ=0 field equations is

$A(t,r) = \gamma\,\dot{a}[t]^2$ where γ is free constant,

$$B(t,r) = a(t)^2\mathbb{B}(r) = a(t)^2\frac{3\,r^2\,\beta\,\gamma\,(2\,\mathbb{C}[r] + r\,\mathbb{C}'[r])^2}{4\left(\gamma + 3\,r^2\,\beta\,\gamma\,\mathbb{C}[r] + 3\,c^2\,r^4\,\beta\,\mathbb{C}[r]^2\right)}$$

where β is a free constant,

$\mathbb{C}(t,r) = a(t)^2\mathbb{C}[r]$ where $\mathbb{C}(r)$ is a free parameter,

$$\mathbb{D}_4(r) = \pm e^{\int_1^r \frac{\pm(2\,\mathbb{C}[\rho]+\rho\,\mathbb{C}'[\rho])}{\rho\,\mathbb{C}[\rho]\,\sqrt{1+3\,\beta\,\rho^2\,\mathbb{C}[\rho]+\frac{3\,c^2\,\beta\,\rho^4\,\mathbb{C}[\rho]^2}{\gamma}}}\,d\rho},$$

$$\mathbb{D}_5(r) = \frac{\pm 1}{\mathbb{D}_4[r]},$$

and line element (using $\mathbb{D}_4(r)$ rather than its formula)

$$ds^2 = \gamma\,\dot{a}[t]^2\,dt^2 - \frac{3\,r^2\,\beta\,\gamma\,a[t]^2\,(2\,\mathbb{C}[r] + r\,\mathbb{C}'[r])^2}{4\left(\gamma + 3\,r^2\,\beta\,\gamma\,\mathbb{C}[r] + 3\,c^2\,r^4\,\beta\,\mathbb{C}[r]^2\right)}\,dr^2 - r^2\,a[t]^2\,\mathbb{C}[r]\,d\theta^2 - p_4^2\,\mathbb{D}_4[r]\,d\psi_4^2$$
$$- r^2\,a[t]^2\,\mathrm{Sin}[\theta]^2\,\mathbb{C}[r]\,d\phi^2$$
$$- \frac{p_5^2}{\mathbb{D}_4[r]}\,d\psi_5^2$$

A simple example of this solution is to set

$$\beta=\gamma=\mathbb{C}[r]=1$$

$$\Rightarrow \quad \mathbb{D}_4(r) = \pm e^{\int_1^r \frac{2}{\rho\,\sqrt{1+3\,\rho^2+3\,c^2\,\rho^4}}\,d\rho}$$

$$= \frac{5\,r^2}{2+3\,r^2},$$

Then

$$A(t,r) = \dot{a}[t]^2,$$

$$\mathbb{B}(r) = \frac{3\,r^2}{1+3\,r^2+3\,c^2\,r^4},$$

$$\mathbb{C}(t,r) = a[t]^2,$$

$$\mathbb{D}_4(r) = \pm\frac{5\,r^2}{2+3\,r^2},$$

and

$$\mathbb{D}_5(r) = \frac{1}{\mathbb{D}_4[r]} = \pm\frac{2+3\,r^2}{5\,r^2}.$$

a-II: Solve $A(t,r)=c^2$ for $a(t)$

Solving $\gamma\,\dot{a}[t]^2 = c^2$ yields

$$a(t) = k + \frac{c\,t}{\sqrt{\gamma}}$$ where k is new, a free constant.

a_II: $B(t,r) = a(t)^2\mathbb{B}(r)$, $\mathbb{C}[t,r] = a[t]^2\mathbb{C}[r]$, $A(t,r)=c^2$

We solve Case a again, but with the extra condition that $A(t,r)=c^2$.

Solution 1 to the $\Lambda=0$ field equations:

Plugging $A(t,r)=c^2$ and $a(t) = k + \frac{c\,t}{\sqrt{\gamma}}$

into the Case a solution gives us one solution to this case, a subcase of Case a:

$$a(t) = k + \frac{c\,t}{\sqrt{\gamma}},$$

$$A(t,r) = c^2,$$

$$B(t,r) = a(t)^2\mathbb{B}(r) = a(t)^2\frac{3\,r^2\,\beta\,\gamma\,(2\,\mathbb{C}[r]+r\,\mathbb{C}'[r])^2}{4\left(\gamma+3\,r^2\,\beta\,\gamma\,\mathbb{C}[r]+3\,c^2\,r^4\,\beta\,\mathbb{C}[r]^2\right)},$$

$$\mathbb{C}(t,r) = a(t)^2\mathbb{C}[r],$$

$$\mathbb{D}_4(r) = e^{\int_1^r \frac{\pm(2\,\mathbb{C}+\rho\,\mathbb{C}')}{\mathbb{C}\,\rho\,\sqrt{1+3\,\mathbb{C}\,\beta\,\rho^2+\frac{3\,c^2\,\mathbb{C}^2\,\beta\,\rho^4}{\gamma}}}\,d\rho}$$

and $\mathbb{D}_5(r) = \frac{1}{\mathbb{D}_4[r]}$.

Begin search for other solutions:

It is possible we could obtain a different solution by setting $A(t,r)=c^2$ before solving the ODEs and, in fact, we find two new solutions below. To generate these solutions, we solve this case from scratch.

After plugging in the formulas for A, B, and $\mathbb{C}$, the field equations become

$$R_{00}: \quad \frac{3\,\ddot{a}}{a} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{a}\,D_4'}{2\,a\,D_4} - \frac{\dot{a}\,D_5'}{2\,a\,D_5} = 0$$

$$R_{11}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{(C')^2}{2\,a^2\,C^2\,B} + \frac{B'}{a^2\,r\,B^2} + \frac{C'\,B'}{2\,a^2\,C\,B^2} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{C''}{a^2\,C\,B} - \frac{D_4''}{2\,a^2\,B\,D_4} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2\,C} - \frac{1}{a^2\,r^2\,B} + \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{B'}{2\,a^2\,r\,B^2} + \frac{C'\,B'}{4\,a^2\,C\,B^2} - \frac{D_4'}{2\,a^2\,r\,B\,D_4} - \frac{C'\,D_4'}{4\,a^2\,C\,B\,D_4} - \frac{D_5'}{2\,a^2\,r\,B\,D_5} - \frac{C'\,D_5'}{4\,a^2\,C\,B\,D_5} - \frac{C''}{2\,a^2\,C\,B} = \Lambda$$

$$R_{44}: \quad -\frac{D_4'}{a^2\,r\,B\,D_4} - \frac{C'\,D_4'}{2\,a^2\,C\,B\,D_4} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} - \frac{D_4'\,D_5'}{4\,a^2\,B\,D_4\,D_5} - \frac{D_4''}{2\,a^2\,B\,D_4} = \Lambda$$

$$R_{55}: \quad -\frac{D_5'}{a^2\,r\,B\,D_5} - \frac{C'\,D_5'}{2\,a^2\,C\,B\,D_5} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} - \frac{D_4'\,D_5'}{4\,a^2\,B\,D_4\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

First, we can solve R_{01} for D_5, obtaining the result above:

$$D_5(r) = \frac{K}{D_4} \quad \text{where } K \text{ is a constant of integration.}$$

Since $g_{55} = -D_5(r)\,p_5{}^2 = \frac{K\,p_5{}^2}{D_4\,(r)}$, we can roll K into the radius p_5:

$$D_5(r) = \pm\frac{1}{D_4}.$$

Next, we can solve R_{00} for $a(t)$:

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\,\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\,\beta \quad \text{where } \alpha \text{ and } \beta \text{ are constants of integration.}$$

Continuing, we can algebraically solve $R_{11}-2R_{22}$ for B:

$$R_{11}-2R_{22}: \quad -\frac{2}{a^2\,r^2\,C} + \frac{2}{a^2\,r^2\,B} - \frac{2\,\dot{a}^2}{a^2} - \frac{\ddot{a}}{a} + \frac{2\,C'}{a^2\,r\,C\,B} + \frac{(C')^2}{2\,a^2\,C^2\,B} - \frac{(D_4')^2}{2\,a^2\,B\,D_4^2} = -\Lambda$$

$$\mathbb{B}(r) = \frac{-4\,\mathbb{C}^2\,\mathbb{D}_4^2 - 4\,r\,\mathbb{C}\,\mathbb{D}_4^2\,\mathbb{C}' - r^2\,\mathbb{D}_4^2\,(\mathbb{C}')^2 + r^2\,\mathbb{C}^2\,(\mathbb{D}_4')^2}{2\,\mathbb{C}\left(-2 + a^2\,r^2\,\mathbb{C}\,\Lambda - 2\,r^2\,\mathbb{C}\,\dot{a}^2 - a\,r^2\,\mathbb{C}\,\ddot{a}\right)\mathbb{D}_4^2}$$

There are no t terms in the numerator of $\mathbb{B}(r)$. In order for $\mathbb{B}(r)$ to be strictly a function of r, the term in parenthesis in the denominator must equal a function of r. Plugging in the formula for $a(t)$, we see that this is true:

$$\text{Parentheses} = -2 + a^2\,r^2\,\mathbb{C}\,\Lambda - 2\,r^2\,\mathbb{C}\,\dot{a}^2 - a\,r^2\,\mathbb{C}\,\ddot{a}$$

$$= -2 + \frac{8}{3}\,r^2\,\alpha\,\beta\,\Lambda\,\mathbb{C}[r], \text{ a function of r.}$$

$$\Rightarrow \mathbb{B}(r) = \frac{-4\,\mathbb{C}^2\,\mathbb{D}_4^2 - 4\,r\,\mathbb{C}\,\mathbb{D}_4^2\,\mathbb{C}' - r^2\,\mathbb{D}_4^2\,(\mathbb{C}')^2 + r^2\,\mathbb{C}^2\,(\mathbb{D}_4')^2}{-4\,\mathbb{C}\,\mathbb{D}_4^2 + \frac{16}{3}\,r^2\,\mathbb{C}^2\,\alpha\,\beta\,\Lambda\,\mathbb{D}_4^2}$$

Plugging this in, the remaining equations are a page long each, too lengthy to display, but $R_{11}-2R_{22}+R_{44}+R_{55}$ simplifies nicely:

$$R_{11}-2R_{22}+R_{44}+R_{55}: \quad -\frac{8\,\alpha\,\beta\,\Lambda + 6\,a'[t]^2 + 3\,a[t]\,a''[t]}{3\,a[t]^2} = \Lambda$$

When we plug the formula for $a(t)$ back in, it simplifies even more:

$$R_{11}-2R_{22}+R_{44}+R_{55}: \quad -\Lambda = \Lambda$$

$$\Rightarrow \quad \Lambda = 0$$

$$\Rightarrow \quad a(t) = \alpha + \beta \equiv \zeta, \text{ a constant.}$$

Because $a[t]$ is only used in two places,

$$\mathbb{B}(t,r) = a[t]^2\mathbb{B}(r) = \zeta\,\mathbb{B}(r) \equiv \hat{\mathbb{B}}(r)$$

and

$$\mathbb{C}(t,r) = a[t]^2\mathbb{C}(r) = \zeta\,\mathbb{C}(r) \equiv \hat{\mathbb{C}}(r),$$

we can just roll ζ into each of $\mathbb{B}(r)$ and $\mathbb{C}(r)$, effectively setting ζ=1:

$$\Rightarrow \quad \mathbb{B}(t,r) = \mathbb{B}(r),$$

$$\Rightarrow \quad \mathbb{C}(t,r) = \mathbb{C}(r),$$

$$\Rightarrow \quad a(t) = 1.$$

Plugging these values back in, the equations are lengthy, but we see that R_{55} and R_{11} are redundant, the same as R_{44} and R_{22}, respectively. This leaves us with two equations in the two unknowns, $\mathbb{C}$ and $\mathbb{D}_4$.

Moreover, since the RHS's are now Λ=0, we can put the two remaining LHS's over common denominators and then set the numerators equal to zero:

R_{22}: $-8\,C^3\,D_4^2\,(D_4')^2 - 4\,r\,C^2\,D_4^2\,C'\,(D_4')^2 - 3\,r^2\,C\,D_4^2\,(C')^2\,(D_4')^2 + 4\,r\,C^3\,D_4\,(D_4')^3 + 2\,r^2\,C^2\,D_4\,C'\,(D_4')^3 + r^2\,C^3\,(D_4')^4 + 2\,r^2\,C^2\,D_4^2\,(D_4')^2\,C'' - 4\,r\,C^3\,D_4^2\,D_4'\,D_4'' - 2\,r^2\,C^2\,D_4^2\,C'\,D_4'\,D_4'' = 0$

R_{44}: $-16\,C^3\,D_4^3\,D_4' - 16\,r\,C^2\,D_4^3\,C'\,D_4' - 10\,r^2\,C\,D_4^3\,(C')^2\,D_4' - 3\,r^3\,D_4^3\,(C')^3\,D_4' + 8\,r\,C^3\,D_4^2\,(D_4')^2 + 8\,r^2\,C^2\,D_4^2\,C'\,(D_4')^2 + 2\,r^3\,C\,D_4^2\,(C')^2\,(D_4')^2 + 2\,r^2\,C^3\,D_4\,(D_4')^3 + r^3\,C^2\,D_4\,C'\,(D_4')^3 + 4\,r^2\,C^2\,D_4^3\,D_4'\,C'' + 2\,r^3\,C\,D_4^3\,C'\,D_4'\,C'' - 8\,r\,C^3\,D_4^3\,D_4'' - 8\,r^2\,C^2\,D_4^3\,C'\,D_4'' - 2\,r^3\,C\,D_4^3\,(C')^2\,D_4'' = 0$

R_{22} can be solved for D_4, and there are 2 solutions (that we label Solution 2 and Solution 3):

Solution 2: $D_4(r) = \gamma$ where γ is a free constant

Solution 3: $D_4(r) = e^{\int_1^r \frac{2\,C+\rho\,C'}{C\,\rho\,\sqrt{1+C\,\gamma\,\rho^2}}\,d\rho}\,\chi$ where γ and χ are free constants.

This leaves just one equation, R_{44}, in one unknown, $D_4(r)$.

Moreover, in each case, $B(r)$ can be simplified by plugging in $D_4(r)$.

Solution 2: $D_4(r) = \gamma$

With this value for $D_4(r)$, the last equation, R_{44}, is satisfied:

R_{44}: $0 = 0$

We can 'roll γ into p_4'. That is, note that $g_{44} = -D_4(r)\,p_4{}^2 = -\frac{D_4(r)}{\gamma}\,(\gamma p_4{}^2) \equiv \pm\hat{D}_4(r)\,(\hat{p}_4{}^2)$. Let D_4 and p_4 be replaced by $\hat{D}_4$ and $\hat{p}_4$, respectively. Then $\hat{D}_4$ represents a valid metric whose formula is

$$\hat{D}_4(r) = \pm\frac{D_4(r)}{\gamma} = \pm\frac{\gamma}{\gamma} = \pm 1.$$

We could continue to solve these equations using $\hat{D}_4$ and $\hat{p}_4$ and, when finished, we could rename $\hat{D}_4$ and $\hat{p}_4$ back to D_4 and p_4. However, we prefer to do the renaming now. Thus:

$$D_4(r) = \pm 1.$$

This process 'effectively' sets the γ term in D_4 to ± 1, and we can similarly set

$$D_5(r) = \pm 1.$$

Plugging in $D_4(r)=1$ simplifies $B(r)$:

$$B(r) = \frac{(2\,C[r] + r\,C'[r])^2}{4\,C[r]}$$

Since all of the equations have been satisfied, we have an exact solution to the $\Lambda=0$ field equations, and it is neither a subcase nor a generalization of Solution 1 because of $a(t)$:

The Solution 1 $a(t) = k + \frac{c\,t}{\sqrt{\gamma}}$ is a function of t and there is no value of γ that makes $a(t)$ constant.

This Solution 2 $a(t)$ is a constant, and is not a function of t.

This solution is:

$a(t) = 1,$

$A(t,r) = c^2,$

$B(t,r) = a(t)^2 B(r) = \frac{(2\,C[r] + r\,C'[r])^2}{4\,C[r]}$

$\mathbb{C}(t,r) = a(t)^2 C(r),$ where $C(r)$ is a free parameter,

$D_4(r) = \pm 1,$

$D_5(r) = \pm 1,$

and line element

$$ds^2 = c^2\,dt^2 - \frac{(2\,C[r] + r\,C'[r])^2}{4\,C[r]}\,dr^2 - r^2\,C[r]\,d\theta^2 - r^2\,\mathrm{Sin}[\theta]^2\,C[r]\,d\phi^2 - p_4^2\,d\psi_4^2 - p_5^2\,d\psi_5^2$$

Solution 3: $D_4(r) = e^{\int_1^r \frac{2\,C+\rho\,C'}{C\,\rho\,\sqrt{1+C\,\gamma\,\rho^2}}\,d\rho}\,\chi$

Plugging this formula for $D_4(r)$ into $B(r)$, the integral terms all cancel out and $B(r)$ can be simplified to

$$B(r) = \frac{r^2\,\gamma\,(2\,C + r\,C')^2}{4 + 4\,r^2\,C\,\gamma}$$

In addition, the last equation, R_{44}, is again satisfied, giving us another solution:

R_{44}: $0 = 0$

As usual, we can roll χ into p_4, so that

$$D_4(r) = \pm e^{\int_1^r \frac{2\,C+\rho\,C'}{C\,\rho\,\sqrt{1+C\,\gamma\,\rho^2}}\,d\rho}.$$

This exact solution to the Λ=0 field equations is neither a subcase nor a generalization of either of the first two solutions:

- As with Solution 2, because $a(t)$ is a constant, this solution is neither a subcase nor a generalization of Solution 1.

- $\mathcal{D}_4(r)=1$ in Solution 2 will be a subcase of the $\mathcal{D}_4$ integral formula of Solution 3 only if the integral equals zero. This requires the numerator of the integrand to be zero, which requires $\mathcal{C}(r) = \frac{k}{r}$ for some constant k. But this means that $\mathcal{C}(r)$ of Solution 3 is a subcase of the free $\mathcal{C}(r)$ of Solution 2. So, neither is a subcase of the other.

Solution 3 consists of:

$$a(t) = 1,$$

$$A(t,r) = c^2,$$

$$B(t,r) = a(t)^2 \mathcal{B}(r) = \frac{r^2 \gamma (2\,\mathcal{C}[r] + r\,\mathcal{C}'[r])^2}{4 + 4\,r^2 \gamma\,\mathcal{C}[r]} \quad \text{where } \gamma \text{ is a free constant,}$$

$$\mathbb{C}(t,r) = a(t)^2 \mathcal{C}(r) = \mathcal{C}[r], \quad \text{a free parameter,}$$

$$\mathcal{D}_4(r) = \pm e^{\int_1^r \frac{2\,\mathcal{C}[\rho]+\rho\,\mathcal{C}'[\rho]}{\rho\,\mathcal{C}[\rho]\,\sqrt{1+\gamma\,\rho^2\,\mathcal{C}[\rho]}}\,d\rho},$$

$$\mathcal{D}_5(r) = \frac{\pm 1}{\mathcal{D}_4(r)},$$

and line element

$$ds^2 = c^2\,dt^2 - \frac{r^2 \gamma (2\,\mathcal{C}[r] + r\,\mathcal{C}'[r])^2}{4 + 4\,r^2 \gamma\,\mathcal{C}[r]}\,dr^2 - r^2\,\mathcal{C}[r]\,d\theta^2 - r^2\,\mathrm{Sin}[\theta]^2\,\mathcal{C}[r]\,d\phi^2$$
$$- e^{\int_1^r \frac{2\,\mathcal{C}[\rho]+\rho\,\mathcal{C}'[\rho]}{\rho\,\mathcal{C}[\rho]\,\sqrt{1+\gamma\,\rho^2\,\mathcal{C}[\rho]}}\,d\rho}\,p_4^2\,d\psi_4^2$$
$$- e^{-\int_1^r \frac{2\,\mathcal{C}[\rho]+\rho\,\mathcal{C}'[\rho]}{\rho\,\mathcal{C}[\rho]\,\sqrt{1+\gamma\,\rho^2\,\mathcal{C}[\rho]}}\,d\rho}\,p_5^2\,d\psi_5^2$$

Example: $\gamma=-1$ and $\mathcal{C}(r)=r^{-4}$:

$$\mathcal{B}(r) = \frac{1}{r^4 - r^6}$$

$$\mathcal{D}_4(r) = \boxed{\frac{1}{\left(r + \sqrt{-1 + r^2}\right)^2} \quad \text{if } \mathrm{Re}[r] > 1 \,\&\&\, \mathrm{Im}[r] == 0}$$

b: $\mathcal{B}(t,r)=\mathcal{B}(r)$, $\mathbb{C}(t,r)=\mathcal{C}(r)$

The field equations become:

$$R_{00}: \quad -\frac{A'}{A\,r\,\mathcal{B}} + \frac{(A')^2}{4\,A^2\,\mathcal{B}} - \frac{A'\,\mathcal{C}'}{2\,A\,\mathcal{C}\,\mathcal{B}} + \frac{A'\,\mathcal{B}'}{4\,A\,\mathcal{B}^2} - \frac{A'\,\mathcal{D}_4'}{4\,A\,\mathcal{B}\,\mathcal{D}_4} - \frac{A'\,\mathcal{D}_5'}{4\,A\,\mathcal{B}\,\mathcal{D}_5} - \frac{A''}{2\,A\,\mathcal{B}} = \Lambda$$

$$R_{11}: \quad \frac{(A')^2}{4\,A^2\,\mathcal{B}} - \frac{2\,\mathcal{C}'}{r\,\mathcal{C}\,\mathcal{B}} + \frac{(\mathcal{C}')^2}{2\,\mathcal{C}^2\,\mathcal{B}} + \frac{\mathcal{B}'}{r\,\mathcal{B}^2} + \frac{A'\,\mathcal{B}'}{4\,A\,\mathcal{B}^2} + \frac{\mathcal{C}'\,\mathcal{B}'}{2\,\mathcal{C}\,\mathcal{B}^2} +$$
$$\frac{\mathcal{B}'\,\mathcal{D}_4'}{4\,\mathcal{B}^2\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4\,\mathcal{B}\,\mathcal{D}_4^2} + \frac{\mathcal{B}'\,\mathcal{D}_5'}{4\,\mathcal{B}^2\,\mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4\,\mathcal{B}\,\mathcal{D}_5^2} - \frac{A''}{2\,A\,\mathcal{B}} - \frac{\mathcal{C}''}{\mathcal{C}\,\mathcal{B}} - \frac{\mathcal{D}_4''}{2\,\mathcal{B}\,\mathcal{D}_4} - \frac{\mathcal{D}_5''}{2\,\mathcal{B}\,\mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2\,C} - \frac{1}{r^2\,B} - \frac{A'}{2\,A\,r\,B} - \frac{2\,C'}{r\,C\,B} - \frac{A'\,C'}{4\,A\,C\,B} + \frac{B'}{2\,r\,B^2} + \frac{C'\,B'}{4\,C\,B^2} - \frac{D_4'}{2\,r\,B\,D_4} - \frac{C'\,D_4'}{4\,C\,B\,D_4} - \frac{D_5'}{2\,r\,B\,D_5} - \frac{C'\,D_5'}{4\,C\,B\,D_5} - \frac{C''}{2\,C\,B} = \Lambda$$

$$R_{44}: \quad -\frac{D_4'}{r\,B\,D_4} - \frac{A'\,D_4'}{4\,A\,B\,D_4} - \frac{C'\,D_4'}{2\,C\,B\,D_4} + \frac{B'\,D_4'}{4\,B^2\,D_4} + \frac{(D_4')^2}{4\,B\,D_4^2} - \frac{D_4'\,D_5'}{4\,B\,D_4\,D_5} - \frac{D_4''}{2\,B\,D_4} = \Lambda$$

$$R_{55}: \quad -\frac{D_5'}{r\,B\,D_5} - \frac{A'\,D_5'}{4\,A\,B\,D_5} - \frac{C'\,D_5'}{2\,C\,B\,D_5} + \frac{B'\,D_5'}{4\,B^2\,D_5} - \frac{D_4'\,D_5'}{4\,B\,D_4\,D_5} + \frac{(D_5')^2}{4\,B\,D_5^2} - \frac{D_5''}{2\,B\,D_5} = \Lambda$$

We note that R_{01} has disappeared because it is the identity 0=0.

We can perform all the algebraic manipulations tried in Case 5D, and more, but, similarly, they are not productive.

Thus, like 5D, this case is too difficult for me to solve. The next step is to impose additional assumptions:

b_Iα: $\mathbf{A(t,r) = a[t]^2}$

b_II: $\mathbf{A(t,r) = c^2}$

bii: $\mathbf{B(r)} = B(r)$, $C(r) = \mathbf{1}$

biii: $\mathbf{B(r) = 1}$, $C(r) = \mathbf{1}$ (if necessary)

To understand how subcases of A(t,r) are labeled and organized, recall cases I and II:

I: $A(t,r) = \mathcal{A}(t)A(r)$

II: $A(t,r) = c^2$

Case I normally has subcases

Iα: $A(t,r) = \mathcal{A}(t)$

Iβ: $A(t,r) = A(r)$.

Here, we are in notebook 6D_1C, Case b where Case b: B(t,r)=B(r) & $\mathbb{C}(t,r)=C(r)$ and Case C: $\mathbb{D}_k(t,r)=D_k$. Because none of these is defined in terms of a(t), we are free to use a(t) rather than $\mathcal{A}(t)$ in subcase Iα.

Also, I have chosen to skip cases I and Iβ because those cases were found too difficult to solve in the 5D notebook, and I would expect them to be even more complex here in 6D.

As for Cases bii and biii, they are standard subcases of Case b. We seldomly find that Case bi brings new solutions.

We work each case listed above from scratch.

b_Iα: B(t,r)=B(r), C(t,r)=C(r), A(t,r)=a(t)^2

Restarting Case b from scratch using these constraints, the field equations become:

R_{00}: $0 = \Lambda$

R_{11}: $$-\frac{2C'}{rCB}+\frac{(C')^2}{2C^2B}+\frac{B'}{rB^2}+\frac{C'B'}{2CB^2}+\frac{B'D_4'}{4B^2D_4}+\frac{(D_4')^2}{4BD_4^2}+\frac{B'D_5'}{4B^2D_5}+\frac{(D_5')^2}{4BD_5^2}-\frac{C''}{CB}-\frac{D_4''}{2BD_4}-\frac{D_5''}{2BD_5}=\Lambda$$

R_{22}: $$\frac{1}{r^2C}-\frac{1}{r^2B}-\frac{2C'}{rCB}+\frac{B'}{2rB^2}+\frac{C'B'}{4CB^2}-\frac{D_4'}{2rBD_4}-\frac{C'D_4'}{4CBD_4}-\frac{D_5'}{2rBD_5}-\frac{C'D_5'}{4CBD_5}-\frac{C''}{2CB}=\Lambda$$

R_{44}: $$-\frac{D_4'}{rBD_4}-\frac{C'D_4'}{2CBD_4}+\frac{B'D_4'}{4B^2D_4}+\frac{(D_4')^2}{4BD_4^2}-\frac{D_4'D_5'}{4BD_4D_5}-\frac{D_4''}{2BD_4}=\Lambda$$

R_{55}: $$-\frac{D_5'}{rBD_5}-\frac{C'D_5'}{2CBD_5}+\frac{B'D_5'}{4B^2D_5}-\frac{D_4'D_5'}{4BD_4D_5}+\frac{(D_5')^2}{4BD_5^2}-\frac{D_5''}{2BD_5}=\Lambda$$

Again, R_{01} has disappeared because it is the identity 0=0.

From R_{00} we have

$$\Lambda = 0$$

We can algebraically solve $R_{11}-2R_{22}-R_{44}-R_{55}$ for $B(r)$:

$R_{11}-2R_{22}-R_{44}-R_{55}$: $$-\frac{2}{r^2C}+\frac{2}{r^2B}+\frac{2C'}{rCB}+\frac{(C')^2}{2C^2B}+\frac{2D_4'}{rBD_4}+\frac{C'D_4'}{CBD_4}+\frac{2D_5'}{rBD_5}+\frac{C'D_5'}{CBD_5}+\frac{D_4'D_5'}{2BD_4D_5}=0$$

$$B(r) = \frac{1}{4CD_4D_5}\Big(4C^2D_4D_5+4rCD_4D_5C'+r^2D_4D_5(C')^2+4rC^2D_5D_4'+2r^2CD_5C'D_4'+4rC^2D_4D_5'+2r^2CD_4C'D_5'+r^2C^2D_4'D_5'\Big)$$

Plugging back in, all of the equations exceed one page in length. However, R_{22} can be made manageable by putting it over a common denominator and then ignoring the denominator. Nonetheless, the equation is still too complex to solve for any of the remaining unknowns.

Instead of algebraically solving for B, we can Dsolve $R_{11}-2R_{22}-R_{44}-R_{55}$ for D_5:

$$D_5(r) = e^{\int_1^r \frac{-\rho^2 D_4 (C')^2-4C^2(D_4+\rho D_4')+C(4BD_4-2\rho C'(2D_4+\rho D_4'))}{C\rho(2\rho D_4C'+C(4D_4+\rho D_4'))}d\rho}\alpha$$ where α is a constant of integration.

When we plug this back in, the integral disappears from all 4 equations but they are all quite lengthy. After manipulation, we can show R_{44} is redundant, the same as R_{22}. We can also ignore the longest equation, R_{55} because $R_{11}-2R_{22}-R_{44}-R_{55}$ is a dependent set of equations satisfied by $B(r)$. But, the remaining equations are just too difficult for me to solve.

I cannot solve this case.

b_II: B(t,r)=B(r), C(t,r)=C(r), A(t,r)=c²

The field equations become:

R_{00}: $0 = \Lambda$

R_{11}: $$-\frac{2C'}{rCB}+\frac{(C')^2}{2C^2B}+\frac{B'}{rB^2}+\frac{C'B'}{2CB^2}+\frac{B'D_4'}{4B^2D_4}+\frac{(D_4')^2}{4BD_4^2}+\frac{B'D_5'}{4B^2D_5}+\frac{(D_5')^2}{4BD_5^2}-\frac{C''}{CB}-\frac{D_4''}{2BD_4}-\frac{D_5''}{2BD_5}=\Lambda$$

R_{22}: $$\frac{1}{r^2C}-\frac{1}{r^2B}-\frac{2C'}{rCB}+\frac{B'}{2rB^2}+\frac{C'B'}{4CB^2}-\frac{D_4'}{2rBD_4}-\frac{C'D_4'}{4CBD_4}-\frac{D_5'}{2rBD_5}-\frac{C'D_5'}{4CBD_5}-\frac{C''}{2CB}=\Lambda$$

R_{44}: $$-\frac{D_4'}{rBD_4}-\frac{C'D_4'}{2CBD_4}+\frac{B'D_4'}{4B^2D_4}+\frac{(D_4')^2}{4BD_4^2}-\frac{D_4'D_5'}{4BD_4D_5}-\frac{D_4''}{2BD_4}=\Lambda$$

R_{55}: $$-\frac{D_5'}{rBD_5}-\frac{C'D_5'}{2CBD_5}+\frac{B'D_5'}{4B^2D_5}-\frac{D_4'D_5'}{4BD_4D_5}+\frac{(D_5')^2}{4BD_5^2}-\frac{D_5''}{2BD_5}=\Lambda$$

These equations are identical to the field equations in the prior case, b_Iα. So, these equations are too difficult for me to solve.

To understand why the equations are the same, we observe that in the initial 6D_1C equations, $\dot{A}$ is always multiplied by either $\dot{B}$ or $\dot{C}$, both of which are zero. Since A' is zero in both cases, and since A(t,r) only appears in terms having either $\dot{A}$ or A', there is no difference in the equations.

bii: B(t,r)=B(r), C(t,r)=1

The field equations become:

R_{00}: $$-\frac{A'}{ArB}+\frac{(A')^2}{4A^2B}+\frac{A'B'}{4AB^2}-\frac{A'D_4'}{4ABD_4}-\frac{A'D_5'}{4ABD_5}-\frac{A''}{2AB}=\Lambda$$

R_{11}: $$\frac{(A')^2}{4A^2B}+\frac{B'}{rB^2}+\frac{A'B'}{4AB^2}+\frac{B'D_4'}{4B^2D_4}+\frac{(D_4')^2}{4BD_4^2}+\frac{B'D_5'}{4B^2D_5}+\frac{(D_5')^2}{4BD_5^2}-\frac{A''}{2AB}-\frac{D_4''}{2BD_4}-\frac{D_5''}{2BD_5}=\Lambda$$

R_{22}: $$\frac{1}{r^2}-\frac{1}{r^2B}-\frac{A'}{2ArB}+\frac{B'}{2rB^2}-\frac{D_4'}{2rBD_4}-\frac{D_5'}{2rBD_5}=\Lambda$$

R_{44}: $$-\frac{D_4'}{rBD_4}-\frac{A'D_4'}{4ABD_4}+\frac{B'D_4'}{4B^2D_4}+\frac{(D_4')^2}{4BD_4^2}-\frac{D_4'D_5'}{4BD_4D_5}-\frac{D_4''}{2BD_4}=\Lambda$$

R_{55}: $$-\frac{D_5'}{rBD_5}-\frac{A'D_5'}{4ABD_5}+\frac{B'D_5'}{4B^2D_5}-\frac{D_4'D_5'}{4BD_4D_5}+\frac{(D_5')^2}{4BD_5^2}-\frac{D_5''}{2BD_5}=\Lambda$$

Again, note that R_{01} is missing because it is the identity 0=0.

$R_{00}-R_{11}+2R_{22}+R_{44}+R_{55}$ can be solved algebraically for B:

$R_{00}-R_{11}+2R_{22}+R_{44}+R_{55}$: $$\frac{2}{r^2}-\frac{2}{r^2B}-\frac{2A'}{ArB}-\frac{2D_4'}{rBD_4}-\frac{A'D_4'}{2ABD_4}-\frac{2D_5'}{rBD_5}-\frac{A'D_5'}{2ABD_5}-\frac{D_4'D_5'}{2BD_4D_5}=4\Lambda$$

$$B(r)=\frac{-4AD_4D_5-4rD_4D_5A'-4ArD_5D_4'-r^2D_5A'D_4'-4ArD_4D_5'-r^2D_4A'D_5'-Ar^2D_4'D_5'}{4A\left(-1+2r^2\Lambda\right)D_4D_5}$$

$$= \frac{A'\left(-4\,r\,D_4\,D_5 - r^2\,D_5\,D_4' - r^2\,D_4\,D_5'\right)}{4\,A\left(-1+2\,r^2\,\Lambda\right)D_4\,D_5} - \frac{4\,D_4\,(D_5 + r\,D_5') + r\,D_4'\,(4\,D_5 + r\,D_5')}{4\left(-1+2\,r^2\,\Lambda\right)D_4\,D_5}$$

In order for the 1st term on the line above to be strictly a function of r, we must have that $\frac{A'(t,r)}{A(t,r)} = f(r)$ for some function f of r (possibly constant).

Solving $\partial_r A(t,r) = f(r)\,A(t,r)$ yields:

$$A(t,r) = e^{\int_1^r f[\rho]\,d\rho}\,a[t]^2$$

where $a(t)$ is the scale factor.

$$\Rightarrow \quad B(r) = \frac{1}{4\,a^2\left(-1+2\,r^2\,\Lambda\right)D_4\,D_5}\,e^{-\int_1^r f\,d\rho}$$

$$\left(-4\,a^2\,e^{\int_1^r f\,d\rho}\,D_4\,D_5 - 4\,a^2\,e^{\int_1^r f\,d\rho}\,r\,f\,D_4\,D_5 - 4\,a^2\,e^{\int_1^r f\,d\rho}\,r\,D_5\,D_4' - a^2\,e^{\int_1^r f\,d\rho}\,r^2\,f\,D_5\,D_4' - 4\,a^2\,e^{\int_1^r f\,d\rho}\,r\,D_4\,D_5' - a^2\,e^{\int_1^r f\,d\rho}\,r^2\,f\,D_4\,D_5' - a^2\,e^{\int_1^r f\,d\rho}\,r^2\,D_4'\,D_5'\right)$$

$$= -\frac{r\,D_4'\,((4+r\,f)\,D_5 + r\,D_5') + D_4\,(4\,(1+r\,f)\,D_5 + r\,(4+r\,f)\,D_5')}{4\left(-1+2\,r^2\,\Lambda\right)D_4\,D_5}$$

Observe that B is now strictly a function of r since we canceled out the t terms.

Plugging in both $f(r)$ and $B(r)$ makes the equations much more unwieldly even though all of the $a(t)$ terms cancel out.

If we only plug in $f(r)$, the equations are more tractible though still unsolvable:

$$R_{00}: \quad -\frac{f}{r\,B} - \frac{f^2}{4\,B} - \frac{f'}{2\,B} + \frac{f\,B'}{4\,B^2} - \frac{f\,D_4'}{4\,B\,D_4} - \frac{f\,D_5'}{4\,B\,D_5} = \Lambda$$

$$R_{11}: \quad -\frac{f^2}{4\,B} - \frac{f'}{2\,B} + \frac{B'}{r\,B^2} + \frac{f\,B'}{4\,B^2} + \frac{B'\,D_4'}{4\,B^2\,D_4} + \frac{(D_4')^2}{4\,B\,D_4^2} + \frac{B'\,D_5'}{4\,B^2\,D_5} + \frac{(D_5')^2}{4\,B\,D_5^2} - \frac{D_4''}{2\,B\,D_4} - \frac{D_5''}{2\,B\,D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2} - \frac{1}{r^2\,B} - \frac{f}{2\,r\,B} + \frac{B'}{2\,r\,B^2} - \frac{D_4'}{2\,r\,B\,D_4} - \frac{D_5'}{2\,r\,B\,D_5} = \Lambda$$

$$R_{44}: \quad -\frac{D_4'}{r\,B\,D_4} - \frac{f\,D_4'}{4\,B\,D_4} + \frac{B'\,D_4'}{4\,B^2\,D_4} + \frac{(D_4')^2}{4\,B\,D_4^2} - \frac{D_4'\,D_5'}{4\,B\,D_4\,D_5} - \frac{D_4''}{2\,B\,D_4} = \Lambda$$

$$R_{55}: \quad -\frac{D_5'}{r\,B\,D_5} - \frac{f\,D_5'}{4\,B\,D_5} + \frac{B'\,D_5'}{4\,B^2\,D_5} - \frac{D_4'\,D_5'}{4\,B\,D_4\,D_5} + \frac{(D_5')^2}{4\,B\,D_5^2} - \frac{D_5''}{2\,B\,D_5} = \Lambda$$

We can algebraically combine these equations to eliminate either D_4, D_5 or f, or to eliminate all 2nd derivatives to obtain an equation that can be algebraically solved for B. None of these strategies led to a solution of the equations.

Another strategy I tried was to algebraically solve for f:

$R_{00}-R_{11}+R_{44}-R_{55}$: $$-\frac{f}{r\,B}-\frac{B'}{r\,B^2}-\frac{D_4'}{r\,B\,D_4}-\frac{f\,D_4'}{2\,B\,D_4}+\frac{D_5'}{r\,B\,D_5}-\frac{B'\,D_5'}{2\,B^2\,D_5}-\frac{(D_5')^2}{2\,B\,D_5^2}+\frac{D_5''}{B\,D_5}=0$$

or $$-\frac{f}{r}-\frac{B'}{r\,B}-\frac{D_4'}{r\,D_4}-\frac{f\,D_4'}{2\,D_4}+\frac{D_5'}{r\,D_5}-\frac{B'\,D_5'}{2\,B\,D_5}-\frac{(D_5')^2}{2\,D_5^2}+\frac{D_5''}{D_5}=0$$

$$f(r)=\frac{-2\,D_4\,D_5^2\,B'-2\,B\,D_5^2\,D_4'+2\,B\,D_4\,D_5\,D_5'-r\,D_4\,D_5\,B'\,D_5'-r\,B\,D_4\,(D_5')^2+2\,r\,B\,D_4\,D_5\,D_5''}{B\,D_5^2\,(2\,D_4+r\,D_4')}$$

When we plug this back in, R_{00}, R_{11}, and R_{22} become quite lengthy, but the other two can be displayed:

R_{44}: $$-\frac{2\,D_4'}{r\,B\,(2\,D_4+r\,D_4')}+\frac{B'\,D_4'}{B^2\,(2\,D_4+r\,D_4')}+\frac{r\,B'\,(D_4')^2}{4\,B^2\,D_4\,(2\,D_4+r\,D_4')}+\frac{r\,(D_4')^3}{4\,B\,D_4^2\,(2\,D_4+r\,D_4')}-\frac{D_4'\,D_5'}{B\,D_5\,(2\,D_4+r\,D_4')}+\frac{r\,B'\,D_4'\,D_5'}{4\,B^2\,D_5\,(2\,D_4+r\,D_4')}-\frac{r\,(D_4')^2\,D_5'}{4\,B\,D_4\,D_5\,(2\,D_4+r\,D_4')}+\frac{r\,D_4'\,(D_5')^2}{4\,B\,D_5^2\,(2\,D_4+r\,D_4')}-\frac{D_4''}{B\,(2\,D_4+r\,D_4')}-\frac{r\,D_4'\,D_4''}{2\,B\,D_4\,(2\,D_4+r\,D_4')}-\frac{r\,D_4'\,D_5''}{2\,B\,D_5\,(2\,D_4+r\,D_4')}=\Lambda$$

R_{55}: $$-\frac{2\,D_4\,D_5'}{r\,B\,D_5\,(2\,D_4+r\,D_4')}+\frac{D_4\,B'\,D_5'}{B^2\,D_5\,(2\,D_4+r\,D_4')}-\frac{D_4'\,D_5'}{B\,D_5\,(2\,D_4+r\,D_4')}+\frac{r\,B'\,D_4'\,D_5'}{4\,B^2\,D_5\,(2\,D_4+r\,D_4')}-\frac{r\,(D_4')^2\,D_5'}{4\,B\,D_4\,D_5\,(2\,D_4+r\,D_4')}+\frac{r\,D_4\,B'\,(D_5')^2}{4\,B^2\,D_5^2\,(2\,D_4+r\,D_4')}+\frac{r\,D_4'\,(D_5')^2}{4\,B\,D_5^2\,(2\,D_4+r\,D_4')}+\frac{r\,D_4\,(D_5')^3}{4\,B\,D_5^3\,(2\,D_4+r\,D_4')}-\frac{D_4\,D_5''}{B\,D_5\,(2\,D_4+r\,D_4')}-\frac{r\,D_4'\,D_5''}{2\,B\,D_5\,(2\,D_4+r\,D_4')}-\frac{r\,D_4\,D_5'\,D_5''}{2\,B\,D_5^2\,(2\,D_4+r\,D_4')}=\Lambda$$

We can put the LHS of $R_{44}-R_{55}$ over a common denominator, set the numerator to zero, and then solve for B:

$R_{44}-R_{55}$: $$r\,D_4\,D_5\,B'\,(D_5\,D_4'-D_4\,D_5')\,(r\,D_5\,D_4'+D_4\,(4\,D_5+r\,D_5'))+B\left(r^2\,D_5^3\,(D_4')^3-2\,r^2\,D_4\,D_5^3\,D_4'\,D_4''-4\,D_4^2\,D_5^3\,(2\,D_4'+r\,D_4'')+D_4^3\left(-r^2\,(D_5')^3+2\,r^2\,D_5\,D_5'\,D_5''+4\,D_5^2\,(2\,D_5'+r\,D_5'')\right)\right)=0$$

$$B(r)=e^{\int_1^r \frac{-8\,D_4^2\,D_5^3\,D_4'+\rho^2\,D_5^3\,(D_4')^3+8\,D_4^3\,D_5^2\,D_5'-\rho^2\,D_4^3\,(D_5')^3-4\,\rho\,D_4^2\,D_5^3\,D_4''-2\,\rho^2\,D_4\,D_5^3\,D_4'\,D_4''+4\,\rho\,D_4^3\,D_5^2\,D_5''+2\,\rho^2\,D_4^3\,D_5\,D_5'\,D_5''}{\rho\,D_4\,D_5\,(-D_5\,D_4'+D_4\,D_5')\,(4\,D_4\,D_5+\rho\,D_5\,D_4'+\rho\,D_4\,D_5')}\,d\rho}\,\beta$$

where β is a constant of integration.

The resulting equations are complex and the integrals do not disappear.

It can be observed that $R_{44}=R_{55}$, but that is as far as I could get.

I can find no further simplifications. This subscase is too difficult to solve.

biii: B(t,r)=1, C(t,r)=1

The equations are

$$R_{00}: \quad -\frac{A'}{A\,r} + \frac{(A')^2}{4\,A^2} - \frac{A'\,D_4'}{4\,A\,D_4} - \frac{A'\,D_5'}{4\,A\,D_5} - \frac{A''}{2\,A} = \Lambda$$

$$R_{11}: \quad \frac{(A')^2}{4\,A^2} + \frac{(D_4')^2}{4\,D_4^2} + \frac{(D_5')^2}{4\,D_5^2} - \frac{A''}{2\,A} - \frac{D_4''}{2\,D_4} - \frac{D_5''}{2\,D_5} = \Lambda$$

$$R_{22}: \quad -\frac{A'}{2\,A\,r} - \frac{D_4'}{2\,r\,D_4} - \frac{D_5'}{2\,r\,D_5} = \Lambda$$

$$R_{44}: \quad -\frac{D_4'}{r\,D_4} - \frac{A'\,D_4'}{4\,A\,D_4} + \frac{(D_4')^2}{4\,D_4^2} - \frac{D_4'\,D_5'}{4\,D_4\,D_5} - \frac{D_4''}{2\,D_4} = \Lambda$$

$$R_{55}: \quad -\frac{D_5'}{r\,D_5} - \frac{A'\,D_5'}{4\,A\,D_5} - \frac{D_4'\,D_5'}{4\,D_4\,D_5} + \frac{(D_5')^2}{4\,D_5^2} - \frac{D_5''}{2\,D_5} = \Lambda$$

We note that R_{01} disappeared because it is the identity, 0=0.

Solving R_{22} for A yields

$$A(t,r) = \frac{a^2\,e^{-r^2\,\Lambda}}{D_4\,D_5}$$ where the scale factor a(t) is a free parameter.

Plugging this back in yields

$$R_{00}: \quad 3\,\Lambda - r^2\,\Lambda^2 + \frac{D_4'}{r\,D_4} - \frac{r\,\Lambda\,D_4'}{2\,D_4} - \frac{(D_4')^2}{2\,D_4^2} + \frac{D_5'}{r\,D_5} - \frac{r\,\Lambda\,D_5'}{2\,D_5} - \frac{(D_5')^2}{2\,D_5^2} + \frac{D_4''}{2\,D_4} + \frac{D_5''}{2\,D_5} = \Lambda$$

$$R_{11}: \quad \Lambda - r^2\,\Lambda^2 - \frac{r\,\Lambda\,D_4'}{D_4} - \frac{(D_4')^2}{2\,D_4^2} - \frac{r\,\Lambda\,D_5'}{D_5} - \frac{D_4'\,D_5'}{2\,D_4\,D_5} - \frac{(D_5')^2}{2\,D_5^2} = \Lambda$$

$$R_{44}: \quad -\frac{D_4'}{r\,D_4} + \frac{r\,\Lambda\,D_4'}{2\,D_4} + \frac{(D_4')^2}{2\,D_4^2} - \frac{D_4''}{2\,D_4} = \Lambda$$

$$R_{55}: \quad -\frac{D_5'}{r\,D_5} + \frac{r\,\Lambda\,D_5'}{2\,D_5} + \frac{(D_5')^2}{2\,D_5^2} - \frac{D_5''}{2\,D_5} = \Lambda$$

$$\Rightarrow \quad R_{00}+R_{44}+R_{55}: \; 3\,\Lambda - r^2\,\Lambda^2 = 3\,\Lambda$$

$$\Rightarrow \quad \Lambda = 0$$

The remaining equations become

$$R_{00}: \quad \frac{D_4'}{r\,D_4} - \frac{(D_4')^2}{2\,D_4^2} + \frac{D_5'}{r\,D_5} - \frac{(D_5')^2}{2\,D_5^2} + \frac{D_4''}{2\,D_4} + \frac{D_5''}{2\,D_5} = 0$$

$$R_{11}: \quad -\frac{(D_4')^2}{2\,D_4^2} - \frac{D_4'\,D_5'}{2\,D_4\,D_5} - \frac{(D_5')^2}{2\,D_5^2} = 0$$

$$R_{44}: \quad -\frac{D_4'}{r\,D_4} + \frac{(D_4')^2}{2\,D_4^2} - \frac{D_4''}{2\,D_4} = 0$$

$$R_{55}: \quad -\frac{D_5'}{r\,D_5} + \frac{(D_5')^2}{2\,D_5^2} - \frac{D_5''}{2\,D_5} = 0$$

$R_{00}+R_{44}+R_{55}=0$, so the three equations are dependent, and we are free declare R_{00} to be redundant.

In addition, we can solve R_{44} and R_{55} for $\mathcal{D}_4$ and $\mathcal{D}_5$, respectively, eliminating those equations:

$\mathcal{D}_4(r) = e^{-\frac{k}{r}}\alpha$ where α and k are constants of integration

$\mathcal{D}_5(r) = e^{-\frac{\kappa}{r}}\beta$ where β and κ are constants of integration

As usual, we can roll α into the radius p_4 and β into the radius p_5:

$$\mathcal{D}_4(r) = \pm e^{-\frac{k}{r}},$$

and $$\mathcal{D}_5(r) = \pm e^{-\frac{\kappa}{r}}.$$

The remaining equation becomes:

$$R_{11}: \quad -\frac{k^2}{2r^4} - \frac{k\kappa}{2r^4} - \frac{\kappa^2}{2r^4} = 0$$

or $$R_{11}: \quad k^2 + k\kappa + \kappa^2 = 0$$

R_{11} is satisfied if

$$k = -\frac{\kappa}{2} \pm \frac{1}{2} i \sqrt{3}\,\kappa$$

$$\Rightarrow \quad \mathcal{D}_4(r) = \pm e^{\frac{\kappa \pm i\sqrt{3}\kappa}{2r}}$$

$$\Rightarrow \quad \mathcal{D}_5(r) = \pm e^{-\frac{\kappa}{r}}$$

We have another exact solution to the $\Lambda=0$ field equations:

$$A(t,r) = a[t]^2 \frac{e^{-\Lambda r^2}}{\mathcal{D}_4[r]\,\mathcal{D}_5[r]} \quad \text{where } a(t) \text{ free parameter,}$$

$$B(t,r) = 1,$$

$$\mathbb{C}(t,r) = 1,$$

$$\mathcal{D}_4(r) = \pm e^{\frac{\kappa \pm i\sqrt{3}\kappa}{2r}} \quad \text{where } \kappa \text{ is a free constant,}$$

$$\mathcal{D}_5(r) = \pm e^{-\frac{\kappa}{r}},$$

and line element is

$$ds^2 = e^{\frac{\kappa}{r} - \frac{\kappa - i\sqrt{3}\kappa}{2r}} a[t]^2\, dt^2 - 1\, dr^2 - r^2\, d\theta^2 - r^2 \operatorname{Sin}[\theta]^2\, d\phi^2 - e^{\frac{\kappa - i\sqrt{3}\kappa}{2r}} p_4^2\, d\psi_4^2 - e^{-\frac{\kappa}{r}} p_5^2\, d\psi_5^2$$

c: $B(t,r)=B(r)$, $\mathbb{C}(t,r)=a(t)^2$

The field equations become

$$R_{00}: \quad \frac{2 c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{a A^2} - \frac{A'}{A r B} + \frac{(A')^2}{4 A^2 B} + \frac{A' B'}{4 A B^2} - \frac{A' \mathcal{D}_4'}{4 A B \mathcal{D}_4} - \frac{A' \mathcal{D}_5'}{4 A B \mathcal{D}_5} - \frac{A''}{2 A B} = \Lambda$$

$$R_{01}: \quad \frac{2 \dot{a}}{a r} - \frac{\dot{a} A'}{a A} = 0$$

$$R_{11}: \quad \frac{(A')^2}{4 A^2 B} + \frac{B'}{r B^2} + \frac{A' B'}{4 A B^2} + \frac{B' \mathcal{D}_4'}{4 B^2 \mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4 B \mathcal{D}_4^2} + \frac{B' \mathcal{D}_5'}{4 B^2 \mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4 B \mathcal{D}_5^2} - \frac{A''}{2 A B} - \frac{\mathcal{D}_4''}{2 B \mathcal{D}_4} - \frac{\mathcal{D}_5''}{2 B \mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} - \frac{1}{r^2 B} + \frac{c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{A'}{2 A r B} + \frac{B'}{2 r B^2} - \frac{\mathcal{D}_4'}{2 r B \mathcal{D}_4} - \frac{\mathcal{D}_5'}{2 r B \mathcal{D}_5} = \Lambda$$

$$R_{44}: \quad -\frac{\mathcal{D}_4'}{r B \mathcal{D}_4} - \frac{A' \mathcal{D}_4'}{4 A B \mathcal{D}_4} + \frac{B' \mathcal{D}_4'}{4 B^2 \mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4 B \mathcal{D}_4^2} - \frac{\mathcal{D}_4' \mathcal{D}_5'}{4 B \mathcal{D}_4 \mathcal{D}_5} - \frac{\mathcal{D}_4''}{2 B \mathcal{D}_4} = \Lambda$$

$$R_{55}: \quad -\frac{\mathcal{D}_5'}{r B \mathcal{D}_5} - \frac{A' \mathcal{D}_5'}{4 A B \mathcal{D}_5} + \frac{B' \mathcal{D}_5'}{4 B^2 \mathcal{D}_5} - \frac{\mathcal{D}_4' \mathcal{D}_5'}{4 B \mathcal{D}_4 \mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4 B \mathcal{D}_5^2} - \frac{\mathcal{D}_5''}{2 B \mathcal{D}_5} = \Lambda$$

From R_{01}:

$A(t,r) = r^2 \mathcal{A}[t]$ where $\mathcal{A}(t)$ is a new unknown.

This eliminates R_{01}, and the remaining equations become

$$R_{00}: \quad -\frac{2}{r^2 B} + \frac{2 c^2 \ddot{a}}{a r^2 \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{a r^2 \mathcal{A}^2} + \frac{B'}{2 r B^2} - \frac{\mathcal{D}_4'}{2 r B \mathcal{D}_4} - \frac{\mathcal{D}_5'}{2 r B \mathcal{D}_5} = \Lambda$$

$$R_{11}: \quad \frac{3 B'}{2 r B^2} + \frac{B' \mathcal{D}_4'}{4 B^2 \mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4 B \mathcal{D}_4^2} + \frac{B' \mathcal{D}_5'}{4 B^2 \mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4 B \mathcal{D}_5^2} - \frac{\mathcal{D}_4''}{2 B \mathcal{D}_4} - \frac{\mathcal{D}_5''}{2 B \mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} - \frac{2}{r^2 B} + \frac{c^2 \dot{a}^2}{a^2 r^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a r^2 \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a r^2 \mathcal{A}^2} + \frac{B'}{2 r B^2} - \frac{\mathcal{D}_4'}{2 r B \mathcal{D}_4} - \frac{\mathcal{D}_5'}{2 r B \mathcal{D}_5} = \Lambda$$

$$R_{44}: \quad -\frac{3 \mathcal{D}_4'}{2 r B \mathcal{D}_4} + \frac{B' \mathcal{D}_4'}{4 B^2 \mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4 B \mathcal{D}_4^2} - \frac{\mathcal{D}_4' \mathcal{D}_5'}{4 B \mathcal{D}_4 \mathcal{D}_5} - \frac{\mathcal{D}_4''}{2 B \mathcal{D}_4} = \Lambda$$

$$R_{55}: \quad -\frac{3 \mathcal{D}_5'}{2 r B \mathcal{D}_5} + \frac{B' \mathcal{D}_5'}{4 B^2 \mathcal{D}_5} - \frac{\mathcal{D}_4' \mathcal{D}_5'}{4 B \mathcal{D}_4 \mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4 B \mathcal{D}_5^2} - \frac{\mathcal{D}_5''}{2 B \mathcal{D}_5} = \Lambda$$

We can solve for $\mathcal{A}$:

$$R_{00}-R_{22}: \quad -\frac{1}{a} - \frac{c^2 \dot{a}^2}{a \mathcal{A}} + \frac{c^2 \ddot{a}}{\mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 \mathcal{A}^2} = 0$$

$$\Rightarrow \quad \mathcal{A}(t) = \frac{c^2 \dot{a}^2}{-1 + a^2 c^2 k}$$ where k is a constant of integration.

Plugging this back in yields

$$R_{00}: \quad \frac{2 c^2 k}{r^2} - \frac{2}{r^2 B} + \frac{B'}{2 r B^2} - \frac{\mathcal{D}_4'}{2 r B \mathcal{D}_4} - \frac{\mathcal{D}_5'}{2 r B \mathcal{D}_5} = \Lambda$$

$$R_{11}: \quad \frac{3 B'}{2 r B^2} + \frac{B' \mathcal{D}_4'}{4 B^2 \mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4 B \mathcal{D}_4^2} + \frac{B' \mathcal{D}_5'}{4 B^2 \mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4 B \mathcal{D}_5^2} - \frac{\mathcal{D}_4''}{2 B \mathcal{D}_4} - \frac{\mathcal{D}_5''}{2 B \mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad \frac{2 c^2 k}{r^2} - \frac{2}{r^2 B} + \frac{B'}{2 r B^2} - \frac{D_4'}{2 r B D_4} - \frac{D_5'}{2 r B D_5} = \Lambda$$

$$R_{44}: \quad -\frac{3 D_4'}{2 r B D_4} + \frac{B' D_4'}{4 B^2 D_4} + \frac{(D_4')^2}{4 B D_4^2} - \frac{D_4' D_5'}{4 B D_4 D_5} - \frac{D_4''}{2 B D_4} = \Lambda$$

$$R_{55}: \quad -\frac{3 D_5'}{2 r B D_5} + \frac{B' D_5'}{4 B^2 D_5} - \frac{D_4' D_5'}{4 B D_4 D_5} + \frac{(D_5')^2}{4 B D_5^2} - \frac{D_5''}{2 B D_5} = \Lambda$$

R_{00} has become redundant, the same as R_{22}

We can combine the equations such the the 2nd derivatives disappear,
resulting in an equation that can be algebraically solved for B:

$$R_{11}-3R_{22}-R_{44}-R_{55}: \quad -\frac{6 c^2 k}{r^2} + \frac{6}{r^2 B} + \frac{3 D_4'}{r B D_4} + \frac{3 D_5'}{r B D_5} + \frac{D_4' D_5'}{2 B D_4 D_5} = -4 \Lambda$$

$$\Rightarrow \quad B(r) = \frac{12 D_4 D_5 + 6 r D_5 D_4' + 6 r D_4 D_5' + r^2 D_4' D_5'}{4 \left(3 c^2 k - 2 r^2 \Lambda\right) D_4 D_5}$$

However, plugging this in makes the equations
more complex, not simpler, so we discard this attempt.

We cannot solve $3R_{00}+3R_{11}-R_{44}-R_{55}$ for $B(r)$,
similar to the 5D case, due to having D_5 in addition to just D_4.

We can solve R_{22} for $D_5(r)$:

$$D_5(r) = e^{\int_1^r \frac{-4 B D_4 + 4 c^2 k B^2 D_4 - 2 B^2 \Lambda \rho^2 D_4 + \rho D_4 B' - B \rho D_4'}{B \rho D_4} d\rho} \alpha$$ where α is a constant of integration.

Surprisingly, the integral completely
cancels out when we plug this formula back into the equations:

$$R_{11}: \quad \frac{10 c^2 k}{r^2} - \frac{6}{r^2 B} - \frac{4 c^4 k^2 B}{r^2} - 3 \Lambda + 4 c^2 k B \Lambda - r^2 B \Lambda^2 + \frac{5 B'}{2 r B^2} -$$
$$\frac{3 c^2 k B'}{r B} + \frac{3 r \Lambda B'}{2 B} + \frac{(B')^2}{2 B^3} + \frac{2 c^2 k D_4'}{r D_4} - \frac{2 D_4'}{r B D_4} - \frac{r \Lambda D_4'}{D_4} + \frac{B' D_4'}{2 B^2 D_4} - \frac{(D_4')^2}{2 B D_4^2} - \frac{B''}{2 B^2} = \Lambda$$

$$R_{44}: \quad -\frac{c^2 k D_4'}{r D_4} - \frac{D_4'}{2 r B D_4} + \frac{r \Lambda D_4'}{2 D_4} + \frac{(D_4')^2}{2 B D_4^2} - \frac{D_4''}{2 B D_4} = \Lambda$$

$$R_{55}: \quad \frac{4 c^2 k}{r^2} - \frac{4 c^4 k^2 B}{r^2} + 4 c^2 k B \Lambda - r^2 B \Lambda^2 - \frac{B'}{2 r B^2} - \frac{3 c^2 k B'}{r B} +$$
$$\frac{3 r \Lambda B'}{2 B} + \frac{(B')^2}{2 B^3} + \frac{c^2 k D_4'}{r D_4} + \frac{D_4'}{2 r B D_4} - \frac{r \Lambda D_4'}{2 D_4} - \frac{(D_4')^2}{2 B D_4^2} - \frac{B''}{2 B^2} + \frac{D_4''}{2 B D_4} = \Lambda$$

Finally, we can algebraically solve R_{44} For B,

$$\mathbb{B}(r) = \frac{\mathbb{D}_4\,\mathbb{D}_4' - r\,(\mathbb{D}_4')^2 + r\,\mathbb{D}_4\,\mathbb{D}_4''}{\mathbb{D}_4\left(-2\,r\,\Lambda\,\mathbb{D}_4 - 2\,c^2\,k\,\mathbb{D}_4' + r^2\,\Lambda\,\mathbb{D}_4'\right)}$$

When we plug in $\mathbb{B}(r)$, we are down to just R_{11} and R_{55}, and one unknown, $\mathbb{D}_4(r)$. Unfortunately, each equation has approximately 100 complex terms, too long to print here. Fortunately, R_{11}-R_{55} has only 22 complex terms, and the denominators all have a common complex factor term that can be removed since R_{11}-R_{55}=0. This results in:

R_{11}-R_{55}:

$$-6\,c^2\,k\,(\mathbb{D}_4')^4 + 2\,r^2\,\Lambda\,(\mathbb{D}_4')^4 - 2\,c^2\,k\,r\,(\mathbb{D}_4')^3\,\mathbb{D}_4'' + r^3\,\Lambda\,(\mathbb{D}_4')^3\,\mathbb{D}_4'' -$$
$$c^2\,k\,r\,\mathbb{D}_4'''\,(\mathbb{D}_4')^2\,\mathbb{D}_4[r] + \frac{1}{2}\,r^3\,\Lambda\,\mathbb{D}_4'''\,(\mathbb{D}_4')^2\,\mathbb{D}_4[r] - \frac{12\,c^2\,k\,(\mathbb{D}_4')^3\,\mathbb{D}_4[r]}{r} - 5\,r\,\Lambda\,(\mathbb{D}_4')^3\,\mathbb{D}_4[r] +$$
$$c^2\,k\,(\mathbb{D}_4')^2\,\mathbb{D}_4''\,\mathbb{D}_4[r] + \frac{1}{2}\,r^2\,\Lambda\,(\mathbb{D}_4')^2\,\mathbb{D}_4''\,\mathbb{D}_4[r] + 3\,c^2\,k\,r\,\mathbb{D}_4'\,(\mathbb{D}_4'')^2\,\mathbb{D}_4[r] -$$
$$\frac{3}{2}\,r^3\,\Lambda\,\mathbb{D}_4'\,(\mathbb{D}_4'')^2\,\mathbb{D}_4[r] - 6\,c^2\,k\,\mathbb{D}_4'''\,\mathbb{D}_4'\,\mathbb{D}_4[r]^2 + 2\,r^2\,\Lambda\,\mathbb{D}_4'''\,\mathbb{D}_4'\,\mathbb{D}_4[r]^2 + \frac{18\,c^2\,k\,(\mathbb{D}_4')^2\,\mathbb{D}_4[r]^2}{r^2} -$$
$$15\,\Lambda\,(\mathbb{D}_4')^2\,\mathbb{D}_4[r]^2 + \frac{18\,c^2\,k\,\mathbb{D}_4'\,\mathbb{D}_4''\,\mathbb{D}_4[r]^2}{r} + 8\,r\,\Lambda\,\mathbb{D}_4'\,\mathbb{D}_4''\,\mathbb{D}_4[r]^2 + 12\,c^2\,k\,(\mathbb{D}_4'')^2\,\mathbb{D}_4[r]^2 -$$
$$5\,r^2\,\Lambda\,(\mathbb{D}_4'')^2\,\mathbb{D}_4[r]^2 - 6\,r\,\Lambda\,\mathbb{D}_4'''\,\mathbb{D}_4[r]^3 + \frac{18\,\Lambda\,\mathbb{D}_4'\,\mathbb{D}_4[r]^3}{r} + 6\,\Lambda\,\mathbb{D}_4''\,\mathbb{D}_4[r]^3$$
$$= 0.$$

However, this equation is just too complex to solve for the last unknown.

c_II: $\mathbb{B}(t,r)=\mathbb{B}(r)$, $\mathbb{C}(t,r)=a(t)^2$, $\mathbb{A}(t,r)=c^2$

The field equations now become:

R_{00}: $$\frac{2\,\ddot{a}}{a} = \Lambda$$

R_{11}: $$\frac{\mathbb{B}'}{r\,\mathbb{B}^2} + \frac{\mathbb{B}'\,\mathbb{D}_4'}{4\,\mathbb{B}^2\,\mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4\,\mathbb{B}\,\mathbb{D}_4^2} + \frac{\mathbb{B}'\,\mathbb{D}_5'}{4\,\mathbb{B}^2\,\mathbb{D}_5} + \frac{(\mathbb{D}_5')^2}{4\,\mathbb{B}\,\mathbb{D}_5^2} - \frac{\mathbb{D}_4''}{2\,\mathbb{B}\,\mathbb{D}_4} - \frac{\mathbb{D}_5''}{2\,\mathbb{B}\,\mathbb{D}_5} = \Lambda$$

R_{22}: $$\frac{1}{a^2\,r^2} - \frac{1}{r^2\,\mathbb{B}} + \frac{\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{\mathbb{B}'}{2\,r\,\mathbb{B}^2} - \frac{\mathbb{D}_4'}{2\,r\,\mathbb{B}\,\mathbb{D}_4} - \frac{\mathbb{D}_5'}{2\,r\,\mathbb{B}\,\mathbb{D}_5} = \Lambda$$

R_{44}: $$-\frac{\mathbb{D}_4'}{r\,\mathbb{B}\,\mathbb{D}_4} + \frac{\mathbb{B}'\,\mathbb{D}_4'}{4\,\mathbb{B}^2\,\mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4\,\mathbb{B}\,\mathbb{D}_4^2} - \frac{\mathbb{D}_4'\,\mathbb{D}_5'}{4\,\mathbb{B}\,\mathbb{D}_4\,\mathbb{D}_5} - \frac{\mathbb{D}_4''}{2\,\mathbb{B}\,\mathbb{D}_4} = \Lambda$$

R_{55}: $$-\frac{\mathbb{D}_5'}{r\,\mathbb{B}\,\mathbb{D}_5} + \frac{\mathbb{B}'\,\mathbb{D}_5'}{4\,\mathbb{B}^2\,\mathbb{D}_5} - \frac{\mathbb{D}_4'\,\mathbb{D}_5'}{4\,\mathbb{B}\,\mathbb{D}_4\,\mathbb{D}_5} + \frac{(\mathbb{D}_5')^2}{4\,\mathbb{B}\,\mathbb{D}_5^2} - \frac{\mathbb{D}_5''}{2\,\mathbb{B}\,\mathbb{D}_5} = \Lambda$$

We can solve R_{00} for $a(t)$

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\,\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\,\beta$$ where α and β are constant of integration.

Plugging this back in makes the field equations more complex, not simpler.

$R_{11}-2R_{22}-R_{44}-R_{55}$ can be algebraically solved for B or it can be Dsolved for D_5:

$R_{11}-2R_{22}-R_{44}-R_{55}$: $$-\frac{2}{a^2 r^2}+\frac{2}{r^2 B}-\frac{2\dot{a}^2}{a^2}-\frac{2\ddot{a}}{a}+\frac{2 D_4'}{r B D_4}+\frac{2 D_5'}{r B D_5}+\frac{D_4' D_5'}{2 B D_4 D_5}-3 = \Lambda$$

Solving algebraically for B

$$B(r) = -\frac{a^2\left(4 D_4 D_5+4 r D_5 D_4'+4 r D_4 D_5'+r^2 D_4' D_5'\right)}{2\left(-2+3 a^2 r^2 \Lambda-2 r^2 \dot{a}^2-2 a r^2 \ddot{a}\right) D_4 D_5}$$

In order for $B(r)$ to be strictly a function of r, the t terms must cancel out. The only t term in the numerator is $a(t)^2$. To cancel out t terms, the expression in parenthesis in the denominator must be proportional to $a(t)^2$.

If we plug in the formula, above, for $a(t)$, the parenthesis expression appears unlikely to be proportional to $a(t)^2$ but it is too complex to rule out values of α and β that might have this result.

If we assume proportionality, that is we assume

$$B(r) = \frac{4 D_4 D_5+4 r D_5 D_4'+4 r D_4 D_5'+r^2 D_4' D_5',}{2 K D_4 D_5}$$

R_{55} becomes redundant, the same as R_{44}, but otherwise the equations become too difficult to solve when we plug $B(r)$ back in.

If we don't assume proportionality, the equations become even more difficult to solve.

This approach is too difficult to solve.

Dsolving for D_5

This formula, too, results in unwieldy equations:

$$D_5(r) = e^{\int_1^r -\frac{2\left(2 a^2 D_4-2 B D_4+3 a^2 B \Lambda \rho^2 D_4-2 B \rho^2 \dot{a}^2 D_4-2 a B \rho^2 \ddot{a} D_4+2 a^2 \rho D_4'\right)}{a^2 \rho\left(4 D_4+\rho D_4'\right)} d\rho} \gamma$$

The simplest of the resulting equations is R_{44}, but solving it makes a mess:

R_{44}: $$-\frac{D_4'}{a^2 r\left(4 D_4+r D_4'\right)}-\frac{3 D_4'}{r B\left(4 D_4+r D_4'\right)}+\frac{3 r \Lambda D_4'}{2\left(4 D_4+r D_4'\right)}-\frac{r \dot{a}^2 D_4'}{a^2\left(4 D_4+r D_4'\right)}-\frac{r \ddot{a} D_4'}{a\left(4 D_4+r D_4'\right)}+\frac{B' D_4'}{B^2\left(4 D_4+r D_4'\right)}+\frac{\left(D_4'\right)^2}{B D_4\left(4 D_4+r D_4'\right)}+\frac{r B'\left(D_4'\right)^2}{4 B^2 D_4\left(4 D_4+r D_4'\right)}+\frac{r\left(D_4'\right)^3}{4 B D_4^2\left(4 D_4+r D_4'\right)}-\frac{2 D_4''}{B\left(4 D_4+r D_4'\right)}-\frac{r D_4' D_4''}{2 B D_4\left(4 D_4+r D_4'\right)} = \Lambda$$

$$\mathbb{B}(r) = \left(e^{\int_1^r \frac{12 a^2 \mathcal{D}_4^2 \mathcal{D}_4'[\sigma] - 4 a^2 \sigma \mathcal{D}_4 \mathcal{D}_4'[\sigma]^2 - a^2 \sigma^2 \mathcal{D}_4'[\sigma]^3 + 8 a^2 \sigma \mathcal{D}_4^2 \mathcal{D}_4''[\sigma] + 2 a^2 \sigma^2 \mathcal{D}_4 \mathcal{D}_4'[\sigma] \mathcal{D}_4''[\sigma]}{a^2 \sigma \mathcal{D}_4 \mathcal{D}_4'[\sigma] (4 \mathcal{D}_4 + \sigma \mathcal{D}_4'[\sigma])} d\sigma} \Big/ \right.$$

$$\left. \left(\mathbb{C}_1 - \int_1^r \frac{1}{a^2 \rho \mathcal{D}_4 \mathcal{D}_4' (4 \mathcal{D}_4 + \rho \mathcal{D}_4')} e^{\int_1^\rho \frac{12 a^2 \mathcal{D}_4^2 \mathcal{D}_4'[\sigma] - 4 a^2 \sigma \mathcal{D}_4 \mathcal{D}_4'[\sigma]^2 - a^2 \sigma^2 \mathcal{D}_4'[\sigma]^3 + 8 a^2 \sigma \mathcal{D}_4^2 \mathcal{D}_4''[\sigma] + 2 a^2 \sigma^2 \mathcal{D}_4 \mathcal{D}_4'[\sigma] \mathcal{D}_4''[\sigma]}{a^2 \sigma \mathcal{D}_4 \mathcal{D}_4'[\sigma] (4 \mathcal{D}_4 + \sigma \mathcal{D}_4'[\sigma])} d\sigma} \left(16 a^2 \Lambda \rho \mathcal{D}_4^3 + 4 \mathcal{D}_4^2 \mathcal{D}_4' - 2 a^2 \Lambda \rho^2 \mathcal{D}_4^2 \mathcal{D}_4' + 4 \rho^2 \dot{a}^2 \mathcal{D}_4^2 \mathcal{D}_4' + 4 a \rho^2 \ddot{a} \mathcal{D}_4^2 \mathcal{D}_4'\right) d\rho \right) \right)$$

I am unable to solve this case.

d: $B(t,r)=a(t)^2$, $\mathbb{C}(t,r)=C(r)$

The field equations become

$$R_{00}: \quad \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{A'}{a^2 A r} + \frac{(A')^2}{4 a^2 A^2} - \frac{A' C'}{2 a^2 A C} - \frac{A' \mathcal{D}_4'}{4 a^2 A \mathcal{D}_4} - \frac{A' \mathcal{D}_5'}{4 a^2 A \mathcal{D}_5} - \frac{A''}{2 a^2 A} = \Lambda$$

$$R_{01}: \quad -\frac{2 \dot{a}}{a r} - \frac{\dot{a} C'}{a C} - \frac{\dot{a} \mathcal{D}_4'}{2 a \mathcal{D}_4} - \frac{\dot{a} \mathcal{D}_5'}{2 a \mathcal{D}_5} = 0$$

$$R_{11}: \quad \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} + \frac{(A')^2}{4 a^2 A^2} - \frac{2 C'}{a^2 r C} + \frac{(C')^2}{2 a^2 C^2} + \frac{(\mathcal{D}_4')^2}{4 a^2 \mathcal{D}_4^2} + \frac{(\mathcal{D}_5')^2}{4 a^2 \mathcal{D}_5^2} - \frac{A''}{2 a^2 A} - \frac{C''}{a^2 C} - \frac{\mathcal{D}_4''}{2 a^2 \mathcal{D}_4} - \frac{\mathcal{D}_5''}{2 a^2 \mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad -\frac{1}{a^2 r^2} + \frac{1}{r^2 C} - \frac{A'}{2 a^2 A r} - \frac{2 C'}{a^2 r C} - \frac{A' C'}{4 a^2 A C} - \frac{\mathcal{D}_4'}{2 a^2 r \mathcal{D}_4} - \frac{C' \mathcal{D}_4'}{4 a^2 C \mathcal{D}_4} - \frac{\mathcal{D}_5'}{2 a^2 r \mathcal{D}_5} - \frac{C' \mathcal{D}_5'}{4 a^2 C \mathcal{D}_5} - \frac{C''}{2 a^2 C} = \Lambda$$

$$R_{44}: \quad -\frac{\mathcal{D}_4'}{a^2 r \mathcal{D}_4} - \frac{A' \mathcal{D}_4'}{4 a^2 A \mathcal{D}_4} - \frac{C' \mathcal{D}_4'}{2 a^2 C \mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4 a^2 \mathcal{D}_4^2} - \frac{\mathcal{D}_4' \mathcal{D}_5'}{4 a^2 \mathcal{D}_4 \mathcal{D}_5} - \frac{\mathcal{D}_4''}{2 a^2 \mathcal{D}_4} = \Lambda$$

$$R_{55}: \quad -\frac{\mathcal{D}_5'}{a^2 r \mathcal{D}_5} - \frac{A' \mathcal{D}_5'}{4 a^2 A \mathcal{D}_5} - \frac{C' \mathcal{D}_5'}{2 a^2 C \mathcal{D}_5} - \frac{\mathcal{D}_4' \mathcal{D}_5'}{4 a^2 \mathcal{D}_4 \mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4 a^2 \mathcal{D}_5^2} - \frac{\mathcal{D}_5''}{2 a^2 \mathcal{D}_5} = \Lambda$$

R_{01} can be simplified and solved:

$$R_{01}: \quad -\frac{2}{r} - \frac{C'}{C} - \frac{\mathcal{D}_4'}{2 \mathcal{D}_4} - \frac{\mathcal{D}_5'}{2 \mathcal{D}_5} = 0$$

$$\mathcal{D}_5[r] = \frac{k}{r^4 C[r]^2 \mathcal{D}_4[r]}$$ where k is a constant of integration.

This eliminates R_{01}, and the remaining equations become

$$R_{00}: \quad \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} + \frac{(A')^2}{4 a^2 A^2} - \frac{A''}{2 a^2 A} = \Lambda$$

$$R_{11}: \quad -\frac{6}{a^2 r^2} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} + \frac{(A')^2}{4 a^2 A^2} - \frac{6 C'}{a^2 r C} - \frac{3 (C')^2}{2 a^2 C^2} - \frac{2 \mathcal{D}_4'}{a^2 r \mathcal{D}_4} - \frac{C' \mathcal{D}_4'}{a^2 C \mathcal{D}_4} - \frac{(\mathcal{D}_4')^2}{2 a^2 \mathcal{D}_4^2} - \frac{A''}{2 a^2 A} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} + \frac{1}{r^2 C} - \frac{A'}{2 a^2 A r} - \frac{A' C'}{4 a^2 A C} + \frac{(C')^2}{2 a^2 C^2} - \frac{C''}{2 a^2 C} = \Lambda$$

$$R_{44}: \quad -\frac{A' D_4'}{4 a^2 A D_4} + \frac{(D_4')^2}{2 a^2 D_4^2} - \frac{D_4''}{2 a^2 D_4} = \Lambda$$

$$R_{55}: \quad -\frac{2}{a^2 r^2} + \frac{A'}{a^2 A r} + \frac{A' C'}{2 a^2 A C} - \frac{(C')^2}{a^2 C^2} + \frac{A' D_4'}{4 a^2 A D_4} - \frac{(D_4')^2}{2 a^2 D_4^2} + \frac{C''}{a^2 C} + \frac{D_4''}{2 a^2 D_4} = \Lambda$$

We can solve $R_{00}-R_{11}$ for C:

$$R_{00}-R_{11}: \quad \frac{6}{a^2 r^2} + \frac{6 C'}{a^2 r C} + \frac{3 (C')^2}{2 a^2 C^2} + \frac{2 D_4'}{a^2 r D_4} + \frac{C' D_4'}{a^2 C D_4} + \frac{(D_4')^2}{2 a^2 D_4^2} = 0$$

$$\Rightarrow \quad C(r) = \frac{K D_4^{\pm\left(\frac{1}{3} i \left(i+\sqrt{2}\right)\right)}}{r^2}$$, where K is a constant of integration.

We now have 5 equations in 2 unknowns, $A(t,r)$ and $D_5(r)$:

$$R_{00}: \quad \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} + \frac{(A')^2}{4 a^2 A^2} - \frac{A''}{2 a^2 A} = \Lambda$$

$$R_{11}: \quad \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} + \frac{(A')^2}{4 a^2 A^2} - \frac{A''}{2 a^2 A} = \Lambda$$

$$R_{22}: \quad \frac{D_4^{\frac{1}{3}-\frac{i\sqrt{2}}{3}}}{K} + \frac{A' D_4'}{12 a^2 A D_4} - \frac{i A' D_4'}{6 \sqrt{2} a^2 A D_4} - \frac{(D_4')^2}{6 a^2 D_4^2} + \frac{i (D_4')^2}{3 \sqrt{2} a^2 D_4^2} + \frac{D_4''}{6 a^2 D_4} - \frac{i D_4''}{3 \sqrt{2} a^2 D_4} = \Lambda$$

$$R_{44}: \quad -\frac{A' D_4'}{4 a^2 A D_4} + \frac{(D_4')^2}{2 a^2 D_4^2} - \frac{D_4''}{2 a^2 D_4} = \Lambda$$

$$R_{55}: \quad \frac{A' D_4'}{12 a^2 A D_4} + \frac{i A' D_4'}{3 \sqrt{2} a^2 A D_4} - \frac{(D_4')^2}{6 a^2 D_4^2} - \frac{i \sqrt{2} (D_4')^2}{3 a^2 D_4^2} + \frac{D_4''}{6 a^2 D_4} + \frac{i \sqrt{2} D_4''}{3 a^2 D_4} = \Lambda$$

R_{00} is redundant, the same as R_{11}.

We can solve $2R_{22}+R_{44}+R_{55}$ algebraically for D_4:

$$2R_{22}+R_{44}+R_{55}: \quad \frac{2 D_4^{\frac{1}{3}-\frac{i\sqrt{2}}{3}}}{K} = 4 \Lambda$$

$$D_4(r) = 2^{\frac{3 i}{i+\sqrt{2}}} (K \Lambda)^{\frac{3 i}{i+\sqrt{2}}} = (2K\Lambda)^{\frac{3 i}{i+\sqrt{2}}}$$

We are down to 4 equations in a single unknown, $A(t,r)$:

$$R_{11}: \quad \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} + \frac{(A')^2}{4 a^2 A^2} - \frac{A''}{2 a^2 A} = \Lambda$$

$$R_{22}: \quad \frac{\left(2^{\frac{3 i}{i+\sqrt{2}}} (K \Lambda)^{\frac{3 i}{i+\sqrt{2}}}\right)^{\frac{1}{3}-\frac{i\sqrt{2}}{3}}}{K} = \Lambda$$

$$R_{44}: \quad 0 = \Lambda$$

$$R_{55}: \quad 0 = \Lambda$$

From R_{44} and R_{55} we have

$$\Lambda = 0.$$

This implies that $\mathbb{D}_4(r) = 0$, which makes
$\mathbb{D}_5(r)$ undefined (unallowble) and $\mathbb{C}(r) = 0$ (also unallowable).

Case d has no possible solution.

Normally I would next investigate subcases to see if they might
yield solutions. However, in the simpler 5D case, even these subcases
were too difficult to solve. So, I don't bother to try them here in 6D.